



Review paper

Current state of elastography research in multiple anatomical structures and respective phantoms

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ABSTRACT

Purpose: Ultrasound elastography (USE) has gained significant attention in recent years as a non-invasive method for assessing tissue elasticity, offering the potential to improve diagnostic accuracy. With the quick development of this technique, there is a growing demand for calibrated phantoms to validate elastography systems. Therefore, the aim of this review was to provide insights into the current state of elastography research in multiple anatomical structures, which include: liver, breast, prostate, kidneys, blood vessels, heart, eyes, brain, muscles, tendons, and respective phantoms.

Methods: The PubMed, Scopus, and Web of Science databases were searched from January 2019 to May 2024, using specifically selected keywords, to identify relevant studies. Studies that did not use phantoms were not included in this review.

Results: Elastography research uses a wide range of phantoms, mainly for validating innovative approaches and evaluating the performance of existing ones. Phantoms made from gelatin, agar, and poly(vinyl alcohol) cryogel (PVA-C), along with commercially available phantoms, were the most common across the compiled papers.

Conclusion: Overall, an effort in the advancement of new elastography approaches is notorious, but the manufacturing of specialized phantoms that more accurately replicate the properties of living tissue remains a field with significant growth potential.

1. Introduction

Disease progression is often accompanied by alterations in tissue structure and composition, leading to changes in tissue stiffness. Consequently, assessing biomechanical properties presents a way for early diagnosis by evaluating the extent of lesions and the degree of progression of several diseases such as cancer, fibrosis, and cardiovascular disease [1,2].

Clinicians have long relied on palpation to detect lesions, but its effectiveness is limited by factors that include lesion size and depth, thick layers of fat or ascites covering the region of interest, and operator expertise [3]. As a result, there has been significant interest in developing more accurate semi-quantitative and quantitative techniques that can evaluate the elastic properties of tissues.

Ultrasound elastography (USE), firstly described in the 1990 s, is a non-invasive imaging technique that has seen widespread clinical adoption in recent decades. This technique uses ultrasound (US) to measure internal tissue deformations in response to an applied force, providing a means to assess tissue elasticity [4]. Assuming that the tissue is completely elastic, and its deformation does not depend on time, its elasticity can be explained by Hooke's Law:

$$\sigma = \Gamma \cdot \varepsilon \quad (1)$$

where σ represents the stress applied, Γ is the elastic modulus and ε represents the strain.

Depending on the direction of deforming forces, three types of elastic modulus can be distinguished: Young's modulus (E), shear modulus (G), and bulk modulus (K). Young's modulus quantifies the material's

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resistance to deformation along the axis of stress, shear modulus determines the amount of shear strain produced by a force tangential to the direction of displacement, and bulk modulus describes how a material responds to changes in volume due to pressure changes.

Further, a relationship between the Young's modulus, E , and the shear modulus, G , can be established as:

$$E = 2G(1 + \nu) \quad (2)$$

where ν represents the Poisson's ratio. Since biological tissues are nearly incompressible due to their high-water content, the Poisson's ratio is near 0.5 [2]. Hence, for a nearly incompressible isotropic elastic body, the Young's modulus is expressed as in the following equation:

$$E = 3G \quad (3)$$

where E represents the elastic modulus and G the shear modulus.

These elastic moduli also influence the propagation speed of mechanical waves in the medium. US propagation through biological tissue involves both longitudinal and shear waves. In ultrasonic imaging, longitudinal waves are used and their speed, c_L , is determined almost entirely by the tissue's density, ρ , which is approximately 1000 kg m^{-3} , and the bulk modulus, K . Shear waves, cause particles to move perpendicular to the direction of the wave and the speed of propagation, c_s , is determined by ρ and the shear modulus, G :

$$c_s = \sqrt{\frac{G}{\rho}} \quad (4)$$

where c_s represents the shear wave speed, G the shear modulus and ρ is the density.

Using equation 3, shear wave velocity (SWV) can be used to estimate the Young's modulus of the tissue:

$$E = 3\rho c_s^2 \quad (5)$$

where E represents the Young's modulus, c_s represents the shear wave speed and ρ is the density.

The longitudinal wave velocity in soft tissues is approximately 1540 m s^{-1} , with minor variations between different tissues [5]. In turn, shear waves propagate about 1000-fold slower ($1 - 10 \text{ m s}^{-1}$), covering a wider range of values. For this reason, the differences in the SWV can be used to obtain information on the elasticity of different tissues, with good contrast resolution [2,5].

USE uses two main techniques to assess tissue stiffness. Strain imaging provides a qualitative assessment of stiffness by measuring the normal strain response to an applied normal stress, using techniques like strain elastography (SE), also referred to as quasi-static elastography, and acoustic radiation force impulse (ARFI). Shear wave imaging techniques, including one-dimensional transient elastography (TE), point shear wave elastography (pSWE), and two-dimensional shear wave elastography (SWE), quantify tissue elasticity by tracking the propagation of shear waves.

1.1. Tissue-mimicking materials

Tissue-mimicking phantoms are essential tools in elastography since they not only allow medical professionals to familiarize themselves with equipment and protocols in controlled settings while minimizing reliance on animal experiments but also because they represent an important way of testing and validating emerging technologies [6–11]. In fact, for this specific technique, differences between systems have been shown to be up to 20 %-25 % in some cases, making the necessity of having a consistent target to evaluate differences between SWE implementations clear [9].

The primary goal of tissue-mimicking materials (TMMs) is to replicate the properties and responses of human tissue as accurately as

possible. In the case of TMMs used for US techniques, acoustic properties, such as speed of sound and attenuation coefficient, are particularly relevant [6,12]. For that reason, the International Electrotechnical Commission (IEC) Technical Committee 87: Ultrasonics published a series of specification standards. One of these recommends that TMMs for conventional B-mode imaging should mimic soft tissues and possess a speed of sound of 1540 m s^{-1} and an attenuation coefficient of $0.5\text{--}0.7 \text{ dB cm}^{-1} \text{ MHz}^{-1}$ over the frequency range of 2–15 MHz [7,10].

TMMs can be either natural or synthetic and have water, oil, or a combination of both as solvents. The ideal phantom should possess tunable acoustic and mechanical properties, be stable over time, have well-characterized temperature and frequency dependence, and have a reproducible fabrication method [7,8,12].

Since water is the main component of living soft tissue, water-based materials are usually good choices for the fabrication of phantoms. Indeed, water-based materials such as gelatin and agar are widely used due to their acoustic similarity to soft tissues. However, both these materials require the application of strict storage protocols, such as refrigeration, wrapping in cling film, or immersion in antimicrobial solutions, to prevent dehydration and microbial growth. Between the two, there is a preference for agar due to its higher melting point, wider range of Young's modulus, and slightly longer shelf life [6,8,10,11,12]. PVA-based phantoms, while still requiring hydration, possess better stability than the previous two materials, and their acoustic and mechanical properties can be tuned through the variation of the number of freeze–thaw cycles, making it easier to achieve the required elastic properties. Another advantageous feature of this material is its compatibility with both magnetic resonance (MR) and US imaging. However, it's important to note that its fabrication method is quite complex and time-consuming [6,10,12,13].

Other materials such as polyvinyl chloride (PVC) and silicone have been studied as alternatives since they possess prolonged stability over time and possess less demanding storage conditions. Nevertheless, their acoustic properties are less representative, requiring the use of appropriate additives. In addition, probe covers are recommended to prevent interactions with PVC plasticizers. Despite this, PVC still represents a suitable choice for applications where longevity is a priority, such as long-term training programs [6,8].

Adding scatterers to the materials is a key strategy for their US imaging. These particles with diameters of several or tens of micrometers can generate backscattering within the matrix, increasing US brightness and heterogeneity, better mimicking the speckled appearance of human tissue. Common additives include graphite, glass beads, silicon carbide, aluminum oxide, and cellulose [6–8].

Finally, there are also commercial phantoms available, whose goal is often quality control and performance testing. For example, Gammex RMI Ltd developed a condensed milk-based TMM, ATS Labs Inc has a urethane rubber-based TMM, and CIRS Inc uses Zerdine™ TMMs [7,11].

Ultimately, the material choice will depend on the intended application, balancing the ability to replicate biological tissue properties, stability, and practicality to meet specific training and research needs. It is also clear that the ideal phantom has yet to be found, and so continued research into different materials is crucial to support evolving elastography techniques.

1.2. Research objective

With the increasing clinical adoption of elastography, several investigation groups have been optimizing existing techniques and developing innovative technologies to enhance the diagnostic accuracy of several diseases. Validation of these techniques using tissue-mimicking phantoms is crucial to ensure their successful clinical implementation. This systematic review aims to provide a comprehensive overview of the current state of elastography research in multiple anatomical structures of interest (Fig. 1) and respective phantoms.

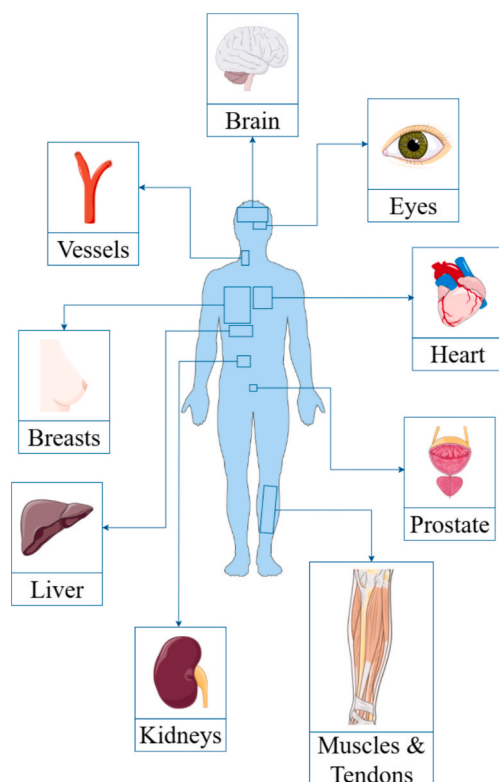


Fig. 1. Schematic representation of the selected anatomical structures of interest encompassed by this study. This figure was designed using images from Servier Medical Art (<https://smart.servier.com>), licensed under a Creative Commons Attribution 4.0 Unported.

2. Methods

A systematic search was conducted on PubMed, Web of Science, and Scopus databases from January 2019 to May 2024. Given the prevalence of elastography studies including the liver, breast, prostate, kidneys, blood vessels, heart, eyes, brain, muscles, and tendons [14–21], these were selected as the anatomical structures of interest for our review. The results of the search were determined based on the title, abstract, and keywords, using the following search terms: (ultrasound OR “ultrasonic wave” OR “acoustic vibration”) AND (“tissue mimicking” OR tissue-mimicking OR phantom OR “tissue substitute” OR “soft tissue”) AND (elastography), followed by the listing of each anatomical structure mentioned above.

All records were extracted to Mendeley Desktop and duplicates were removed using the software filter, followed by a manual review. The titles and abstracts of the retrieved articles were screened to identify relevant studies. The inclusion criteria encompassed articles using tissue phantoms for elastography. Publications were excluded if they were reviews, books, letters to the editor and conference abstracts, were unavailable in English, or did not include phantoms (Fig. 2). A table that contains the authors’ names, publication year, materials, phantom type and purpose, and the elastography technique is presented in each section. A detailed summary of the principal findings from the revised studies is then provided.

3. Results

In this review, 85 studies met all criteria and were included for analysis. These studies have been categorized based on anatomical structure and are presented in the subsequent sections. Fig. 3 summarizes the quantitative values that were measured.

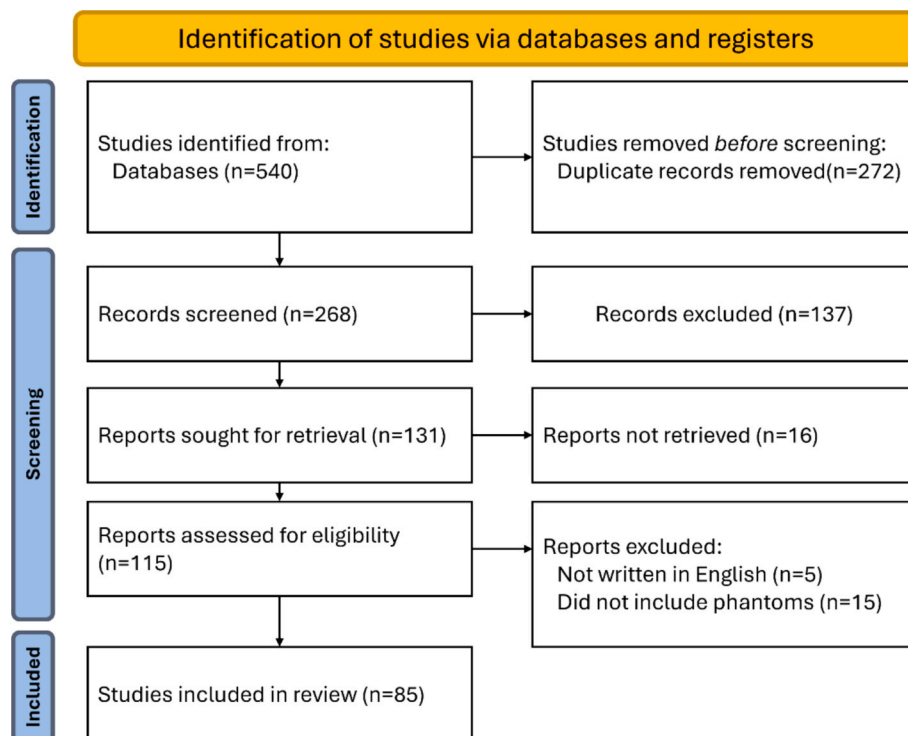


Fig. 2. Diagram representing the process of studies’ screening for this review.

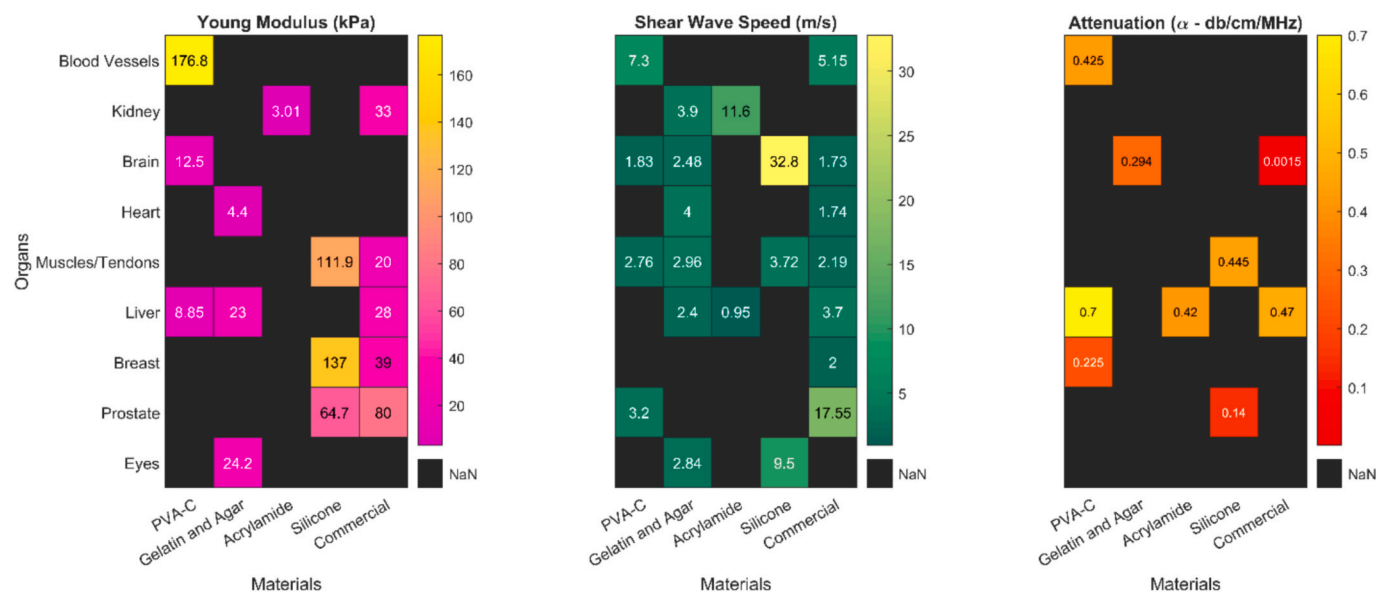


Fig. 3. Heatmap summarizing the reported values of Young’s modulus (kPa), shear wave speed (m/s), and acoustic attenuation (α – dB/cm/MHz) for phantoms used across different organs and materials. NaN indicates values that were either not reported in the original study or not applicable to the specific phantom type.

Table 1
Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for blood vessels in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
PVA-C (10 wt%) + cellulose particles (3 wt %) Water mixture of cornstarch particles (1 wt%) mimicking scattering blood + dissolved salt (5 wt%)	Carotid Artery	Mimic realistic geometries and mechanical properties of the carotid bifurcation whilst being compatible with US imaging	SWE, SE	342 ± 25	11.3 ± 0.1	N/A	Chayer et al., 2019 [25]
PVA-C (10 wt%) + ethanol + cellulose (3 wt %)	Carotid Artery with and without atherosclerosis	Compare the results of the two models and evaluate the addition of time-ensemble and incompressibility constraints to them	2D US elastography with APBE and LSME	N/A	N/A	N/A	Li et al., 2019 [30]
PVA-C (10 wt%) + cellulose (3 wt%)	Vessel with and without lipid plaque	Validate the technique and compare it with the state-of-the-art window-based Lagrangian speckle model estimator	SE with computation using Horn-Schunk optical flow method	182 ± 21	N/A	N/A	Li et al., 2020 [31]
PVA-C (8, 10 wt%) + ethylene glycol (35, 37 wt%) + silica gel particles (0.5, 1.3–2.0 wt%)	Vessel	Evaluate the performance of beam-steered ARFI and wave-tracking for wave velocity estimation along the entire wall circumference compared to traditional 0° transverse ARFI and wave-tracking	SWE	117.5 ± 67.5	5.75 ± 0.75	N/A	Pruijssen et al., 2023 [23]
Zerdine ® (CIRS, Model 040GSE) PVA-C (10 wt%) + ethylene glycol (35.5 wt %) + silica gel particles (1.3 wt%)	Tissue-mimicking Vessel	Compare the performance of different US machines/ probes	SWE	N/A	5.15 ± 0.05	N/A	Pruijssen et al., 2024 [24]
PVA-C (10 %wt) PVA-C (10 %wt) + antibacterial agent PVA-C (10 %wt) + silicon carbide particles	Tissue-mimicking Vessel	Mimicking the properties of healthy and diseased arteries found <i>in vivo</i>	US	79 ± 11	N/A	0.30 ± 0.03	King et al., 2011 [26]
PVA-C (10 %wt) + glycerol (5 %wt) PVA-C (15 %wt) + glycerol (5 %wt)	Tissue-mimicking Vessel	Mimicking the geometry and properties of atherosclerosis blood vessels	US, SWE	164 ± 34	5.46	0.55 ± 0.025	Malone et al., 2020 [27]

3.1. Blood vessels

3.1.1. Motivation

Ischemic stroke is one of the leading causes of death and disability worldwide, with carotid arteriosclerosis accounting for 10–20 % of these cases [22]. Currently, both stroke risk assessment and treatment rely on the degree of stenosis. However, there is growing evidence suggesting that the composition of the plaque may also be correlated to the likelihood of a stroke. Plaques rich in lipids with a thin cap are less rigid and thus more susceptible to rupture, whereas fibrous or calcified plaques, which are stiffer, have a lower risk of rupture [23,24].

Furthermore, a series of other diseases, such as hypertension and Behçet disease, have been associated with higher carotid wall stiffness. It is thus essential to develop or improve non-invasive methods that allow the determination of the mechanical properties of the vessels [24].

US techniques are particularly interesting for this since they are not only non-invasive but also because they have shown their potential for predicting carotid artery plaque instability. It is important to note, however, that, as previously mentioned, these techniques are based on several mechanical assumptions that may not correspond to the truth. For example, it is known that wave propagation in arterial walls differs from propagation in large, homogeneous structures due to their thinness (~0.5–2 mm) [24,25].

For the reasons herein exposed, it is mandatory to find strategies to overcome these limitations. Since all these methods require phantom experiments as a first step validation, it is, then, also clear the need to develop phantoms that have both the geometry and composition of the arteries in mind [25].

3.1.2. Material development

The studies analyzed (Table 1) show a clear concern when it comes to the replication of the geometry of the carotid and the use of several fabrication methods to represent both the artery and any lipid plaque it may have. The most common approach was proposed by Chayer *et al.* [25], who suggested the use of poly(vinyl alcohol) cryogel (PVA-C) with different freeze–thaw cycles to obtain the different mechanical properties desired. In this particular study, the geometry of the mold used was achieved by basing it on a computed tomography scan. The model was digitally modified so that the thickness of the vessel wall was uniformly 2 mm, and it possessed a stenosis of 70 %. PVA-C (10 wt%) used for the lumen underwent six freeze–thaw cycles, while the material used for the soft inclusion (mimicking the lipid plaque responsible for the chosen stenosis level) experienced only one freeze–thaw cycle. The elasticity of the soft inclusion proved itself to be relatively close to the estimated value assessed by other studies (17 ± 3 kPa). The Young's modulus of the PVA-C mimicking the vessel wall was, however, under the minimum values reported in the literature (342 ± 25 kPa, when the range reported for atherosclerotic carotid arteries is between 701 ± 324 kPa to 983 ± 557 kPa). Fundamentally, the use of a higher number of freeze–thaw cycles has, in other studies, proven itself capable of achieving closer values to the desired range. Shear wave speeds clustered between 5–11 m/s, overlapping with the physiological range observed in diseased vessels, thereby underscoring their translational relevance. However, acoustic attenuation was rarely reported, with only a few studies such as King *et al.* [26] and Malone *et al.* [27] quantifying α within the recommended IEC range for vascular tissues. This lack of comprehensive acoustic characterization may limit the comparability of phantoms across studies and their suitability for regulatory validation. Previous studies have also emphasized the importance of detailed viscoelastic characterization of PVA-based phantoms for vascular applications, particularly for Doppler and intravascular US imaging [28,29]. Importantly, while these models succeed in replicating geometry and stiffness contrasts, the relatively narrow focus on isotropic materials overlooks the anisotropy and layered structure of arterial walls, potentially compromising the accuracy of elastography calibration. The proposed fabrication method successfully produced phantoms compatible with US

imaging that accurately replicated the shape and mechanical properties of both normal and stenosed carotid arteries. The material's suitability for this application was further supported by its use in other studies [23,24,30,31].

3.1.3. Technique development

The studies developed by Li *et al.* [30] and Li *et al.* [31], also using PVA-C phantoms, on the other hand, focused on the improvement of the accuracy of SWE for vascular applications by tailoring acquisition and postprocessing methods to these types of applications. Li *et al.* [30] presented two affine model-based estimators: the affine phase-based estimator (APBE) developed in the framework of transverse oscillation beamforming, and the Lagrangian speckle model estimator (LSME). Through the use of atheroma cap phantoms fabricated with 6 freeze–thaw cycles possessing a soft inclusion that only went through 1 of these cycles, it was determined that LSME outperformed the implemented APBE. LSME yielded not only the lowest errors compared to ground truth measurements but also the highest contrast-to-noise ratios for a heterogeneous vascular phantom with a soft inclusion across applied strains ranging from 0.07 % to 3.6 %.

The advancements in knowledge are evident in [31], as the developed sparse model strain estimator (SMSE) reduced estimation errors by up to 50 % compared to the current window-based LSME. The SMSE also demonstrated greater robustness against both global and local noise than the LSME. Simulations, along with *in vitro* and *in vivo* results, indicated a reduction in SMSE processing time by 4 to 25 times compared to the LSME. In addition, the cumulative error assessments performed *in vitro* and *in vivo* showed that residual strains with the SMSE were 2 to 3 times lower than those with the LSME. Furthermore, phantom studies using a PVA-C phantom—fabricated with six freeze–thaw cycles and a soft inclusion representing lipid plaque that underwent only one of these cycles—highlighted the superior spatial resolution of the SMSE algorithm over the LSME.

Pruijssen *et al.* [23] focused on the development of novel technologies, proposing to investigate the potential of beam-steered ARFI and wave-tracking to extend the estimation of group velocity to a larger portion of the arterial circumference compared to conventional 0° ARFI-induced wave-tracking. By utilizing electronic beam-steering, the authors hypothesized that wave velocity estimation could be enabled for additional segments of the transverse arterial cross-section, thus providing a solution for stiffness estimation across a greater proportion of the vessel circumference. Indeed, the application of this technique resulted in a 20 % reduction in the proportion of the circumference with untraceable waves in PVA-C phantoms (8 wt% or 10 wt%) and a 10 % reduction in *ex vivo* experiments, primarily at 0° push locations, which could potentially lead to improved vascular health assessment.

3.1.4. Equipment reliability

Pruijssen *et al.* [24], on the other hand, opted to look at the differences between machines sold by different vendors which may lead to less reliable results. SWE measurements were performed over varying depths on the cylindrical targets of a commercial phantom and in the anterior and posterior walls of a vessel-mimicking phantom (PVA-C, subjected to 2 freeze–thaw cycles). Substantial variability was observed across different US machines, probes, and depths, particularly in the vessel-mimicking phantom. Moreover, higher stiffness values were consistently underestimated, highlighting the limitations of applying the technique to vascular structures.

Despite the limitations mentioned earlier, the potential of these methods to evaluate the mechanical properties of different vessels was clear. In essence, further research into how to improve US imaging is of great interest. The importance of continuing to develop new materials that replicate as many characteristics of the carotid as possible to ensure reliable results was also apparent.

3.2. Kidneys

3.2.1. Motivation

Currently, renal biopsy is considered the gold standard for diagnosing kidney health. However, since this is an invasive method, it cannot be used regularly and may even lead to health complications. For that reason, it's essential to develop non-invasive methods for accurate kidney assessment [32].

Since prolonged kidney inflammation promotes the process of fibrosis and organs with established fibrosis are known to be mechanically stiffer due to the continuous deposition of extracellular matrix, investigating techniques such as elastography, which can assess tissue stiffness, is particularly important [33,34]. Table 2 summarises the studies that have investigated this technique in the last 5 years.

The architecture of the kidney parenchyma is anisotropic, making some of the assumptions inherent to such technique erroneous, making it particularly challenging to rely on this method. In fact, some studies have shown that while SWE has been used to assess kidneys, conflicting results have been obtained [34].

Accordingly, the development of phantoms that are capable of mimicking these particular characteristics is highly relevant. In spite of that, in the last 5 years, there has been a certain lack of concern regarding this aspect and, instead, a greater reliance on commercial or more generic gelatin-based phantoms.

3.2.2. Technique development

There has been a greater interest on the development of new technologies, such as in Kijanka *et al.* [33], which focused on the development of a generalized Stockwell transformation (GST-SFK) for the calculation of dispersion curves in viscoelastic media over an expanded frequency band to better interpret the results obtained through SWE. The robustness of this method was evaluated through the data derived from five custom-made viscoelastic tissue-mimicking phantoms and 60 *in vivo* renal transplants. Through these measurements, it was possible to observe the advantages of the GST-SFK method over the standard two-dimensional Fourier transform (2D-FT) method. In fact, the GST-SFK method exhibited a lower standard deviation in dispersion curve estimates across a broader frequency band compared to the 2D-FT method. These advantages are significant for clinical applications by facilitating the distinction between healthy and non-healthy tissue based on mechanical properties.

A certain interest in the evaluation and optimization of the technique was also noted. Leong *et al.* [34], evaluated the accuracy of SWE-measured stiffness, the effect of tissue anisotropy, and angle of measurement on SWE measurements. For such, the authors used both SWE measurements and dynamic mechanical analysis (DMA) in gelatin phantoms and *ex vivo* sheep kidneys and supplemented the obtained results with an *in vivo* study with healthy volunteers and patients suffering from different levels of interstitial fibrosis. Through these tests, it was possible to conclude that tissue stiffness and anisotropy affected

SWE measurements. Both pSWE and SWE measurements appeared to exhibit a nonlinear relationship with gelatin concentration, even though DMA measurements displayed a fairly linear relationship. Nevertheless, strong correlations were observed between pSWE, SWE, and DMA values, confirming that as gelatin concentration rose, values from both DMA and US techniques also increased, validating the use of these techniques for determining tissue stiffness. The Wilcoxon rank-sum test revealed that median stiffness measurements at 90° were significantly lower than those at 0° across all three methods, suggesting that slight changes in transducer angle to the kidney could substantially affect stiffness readings. This finding underscored the significance of considering the angle of the transducer to the kidney, as slight variations can significantly affect stiffness measurements. The variability of stiffness caused by intra- and inter-operator factors was studied by Kishimoto *et al.* [35]. The SWV of five elasticity phantoms and seven organs- thyroid, lymph node, muscle, spleen, kidney, pancreas, and liver- of 30 healthy volunteers was measured by two different operators. In homogeneous acrylamide-based phantoms, all intraclass correlation coefficients (ICCs) exceeded 0.90, with 95 % Bland-Altman limits of agreement between operators under ± 18 %. While intra-operator ICCs were greater than 0.75 for most organs, except for the pancreas, inter-operator ICC was only above 0.75 for the right hepatic lobe at 4 cm depth and the kidneys, with the 95 % limits of agreement remaining under ± 25 % solely for the liver. This study concluded that, although inter-operator reproducibility was excellent in phantoms, it was insufficient for most anatomical regions in volunteers. These findings highlighted the need for clinicians to consider inter-operator variability when interpreting SWV results.

As noted, the phantoms used in all the mentioned studies are more simplified and, for this reason, demand to be supplemented by results obtained with *ex/in vivo* testing. To minimize this reliance, especially in the early stages of studies, it is thought that there should be more research into the development of phantoms with the capacity to better mimic the kidneys, both healthy and unhealthy. Moreover, as shown in the study of Kishimoto *et al.* [35], the fabrication of phantoms that better mimic the anatomy will improve strategies to train clinicians on how to correctly apply the techniques.

3.3. Brain

3.3.1. Motivation

Several studies have shown a connection between the mechanical properties of the brain and certain chronic diseases, such as Alzheimer's and Parkinson's, and aging. Indeed, aging has been found to affect not only brain size and cognitive function but also brain elasticity, with decreased values of stiffness being correlated to older age [36,37,38].

Consequently, the development of techniques capable of determining the mechanical properties of the brain is a crucial step to improve diagnosis and treatment. Over the past decades, brain diagnosis has been highly dependent on magnetic resonance elastography (MRE)

Table 2

Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for kidneys in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Custom tissue-mimicking phantoms (CIRS) (Material isn't described)	Tissue-mimicking	Evaluate the robustness of the method	SWE with GST-SFK method	33 ± 20	N/A	N/A	Kijanka <i>et al.</i> , 2024 [33]
Gelatin (13.3–53.3 wt %)	Tissue-mimicking	Evaluate how stiffness and tissue anisotropy influence SWE measurements	pSWE, SWE	N/A	3.9 ± 0.2	N/A	Leong <i>et al.</i> , 2020 [34]
Acrylamide + graphite particles	Tissue-mimicking	Verify the reproducibility of SWV measurements with pSWE	pSWE, MRE	3.01 ± 1.6	11.6 ± 7.6	N/A	Kishimoto <i>et al.</i> , 2019 [35]

[37,38]. However, the application of this technique outside of the hospital is particularly challenging, and it is not suitable in emergency cases, such as in the case of intracranial hemorrhage where there is a critical need to expeditiously detect brain injury [39]. Besides, while MRE has been widely used, its spatial resolution limits the accuracy of the imaging results in small regions of interest (ROI) [40].

Medical US appears, then, as a very interesting alternative imaging modality. Despite this, it is essential to bear in mind that the use of US in the brain also comes with several limitations that need to be resolved, namely the fact that in adult humans there's a high acoustic impedance between the skull and brain interface that heavily interferes with US imaging. This, together with the variability in skull geometry and composition between patients, leads to inconsistencies in the obtained results [39,41]. It is also important to note that commercially available phantoms are quite limited in their ability to represent the brain. For instance, the widely used CIRS 049 phantom includes stiff inclusions (up to 80 kPa) that do not accurately represent the typical shear stiffness of brain tissue (1–5 kPa) [36]. It is clear that these types of phantoms are still the preferred selection for studies carried out in the past 5 years [39,42,43]. Alternatively, it was also clear a reliance on generic phantoms [37,38,40,44], which merely mimic soft tissue and so are not

specific to the brain. Table 3 summarises all the studies analyzed.

3.3.2. Material development

A study that stands out was written by McIlvain *et al.* [36], where the factors that affect the production of agar phantoms were systematically analysed to develop a standardized protocol. Although it is not directly stated that the parameters tested are intended to mimic an organ, the article refers to phantoms used specifically to evaluate techniques in the brain, so it is assumed that this was a consideration that was not made explicit. The authors opted to evaluate the effect of the hot plate temperature (300 °C v. 550 °C), salt content (0.3 wt% v. 0.9 wt%), stir rate (100 rpm v. 800 rpm), presence of a lid during heating, final liquid temperature (84 °C v. 96 °C), and the cooling location (room temperature, refrigerator, ice bath). The concentration of agar, was, on the other hand, held constant (1.0 wt%). It was possible to conclude that the phantom homogeneity was affected by both cooling rate and liquid temperature, while phantom stiffness was dependent on the final liquid temperature, salt content in the phantoms, and the interaction of these two metrics with stir rate. Another study that used agar was the one carried out by Zhu *et al.* [40], that proposed an air-coupled US transducer-based optical coherence elastography (AcUT-OCE) technique

Table 3

Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for the brain in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Agar (0.6–1.2 wt%) + salt (0.3–0.9 wt%)	Tissue-mimicking	Evaluate the variation of: agar and salt concentration, hot plate and final liquid temperature, stir rate, whether a lid was used during heating, and cooling location	MRE with nonlinear inversion algorithm	N/A	2.63 ± 0.20	N/A	McIlvain <i>et al.</i> , 2019 [36]
Agar (0.3 % w/v) + lipid solution (0.6 % v/v)	Tissue-mimicking	Verify the feasibility of the proposed protocol	OCE	N/A	2.85 ± 0.12	N/A	Zhu <i>et al.</i> , 2023 [40]
Gelatin (6–15 wt%) + evaporated milk (350 g)	Brain	Understand the bias underlying direct estimation of <i>in situ</i> acoustic intensity using solely MR-ARFI	Combination of MRE and MR-ARFI	N/A	1.84 ± 0.15	0.294 ± 0.03	Li <i>et al.</i> , 2022 [41]
Gelatin (6 wt%) + castor oil (10–30 wt%) + evaporated milk (350 g)							
Zerdine ® (CIRS, Model 073)	Brain	Evaluate repeatability, robustness, and reproducibility of the technique	MRE with guided pressure waves	N/A	1.73 ± 0.14	N/A	Tardieu <i>et al.</i> , 2022 [42]
Bicomponent silicone rubber + softener (1A:1B:2.5S, 1A:1B:1.5S)	Tissue-mimicking	Technique validation	MRE	N/A	32.86 ± 0.14	N/A	Yushchenko <i>et al.</i> , 2022 [45]
Agar (0.6 % w/v) + intralipid solution (0.6 % v/v)	Tissue-mimicking	Evaluate system adequacy and analyze the supershear Rayleigh waves (SRW) behaviors in a homogeneous medium	OCE	N/A	2.72 ± 0.11	N/A	Zhu <i>et al.</i> , 2023 [37]
Zerdine ® (CIRS, Model 037) (Represents the brain) + Quartz container (Represents de skull)	Brain + Skull	Demonstrate the operation of the developed system	RFUS	N/A	N/A	0.0015	Haupt <i>et al.</i> , 2022 [39]
PVA-C (5 wt%) + cellulose (2 wt%)	Tissue-mimicking	Determine accuracy and resolution during HFUSE	HFUSE	12.5 ± 0.75	1.83 ± 0.46	N/A	Lay <i>et al.</i> , 2019 [38]
Gelatin (5, 7 wt%) + graphite (2 wt%)							
Gelatin (5 wt%) + carrageenan (0.1 wt%) + cellulose (1 wt%)	Tissue-mimicking	Technique validation	SWE with SWENet method	N/A	3.38 ± 1.43	N/A	Yin <i>et al.</i> , 2024 [44]
Inclusions: Gelatin (5 wt%) + carrageenan (0.2 wt%) + cellulose (2 wt%)							
Zerdine ® (CIRS, Model 525)	Tissue-mimicking	Technique validation	TE with spatio-temporally smoothed coherence factor) + minimum variance algorithm	N/A	N/A	N/A	Du <i>et al.</i> , 2019 [46]

that allows the noncontact excitation and detection of longitudinal shear waves as well as the assessment of the non-uniformity in brain biomechanical properties. Once again, although the intention to mimic a specific organ was not explicitly stated, the longitudinal shear waves velocity results were very similar to those obtained on the right side of cerebral vessels. An additional interesting study was conducted by Li *et al.* [41]. Its aim was to develop and evaluate a method for accurately estimating acoustic intensity at the focus by utilizing MR-ARFI displacements along with viscoelastic properties obtained with multifrequency MRE. The study used gelatin and castor oil phantoms that demonstrated a range of viscoelastic properties found in human brain tissue. The relaxation rates of ARF-induced displacement responses were found to be between 0.13 and 0.18 m s⁻¹, consistent with previous experimental observations. This was done to better understand the biases associated with the direct estimation of *in situ* acoustic intensity using only MR-ARFI.

However, as noted, none of these studies focused on US technology, which means that the adequacy of the developed phantoms for such techniques isn't assured. As previously mentioned, there is still a heavy reliance on other techniques that are not US-based, such as MR [36,41,42,45] and OCE [37,40]. This is clearly underscored by the fact that such studies were still observed in the systematic research made. Nevertheless, still regarding materials and now focusing on the usage of US, there is the study by Haupt *et al.* [39], that, despite having used a commercial model to mimic the brain, used a quartz container to represent the skull. Thus enhancing the reliability of the developed system. The proposed technique involved the use of pulsed radio frequency energy to generate US waves in brain tissue through thermoelastic expansion. Since conventional US proves ineffective in delivering the high-frequency acoustic waves required for high-resolution brain imaging through the adult skull, it becomes evident that the mimicking of the bone is essential to validate such studies. The authors compared the measured results to FD simulation of the phantom obtaining a good agreement, showing the potential of the developed technique.

3.3.3. Technique development

Other studies that focused on the use of US technology were the ones by Lay *et al.* [38], Yin *et al.* [44] and Du *et al.* [46], which chose to use tissue-mimicking phantoms that don't have the explicit goal of capturing the brain behaviour.

Lay *et al.* [38] proposed a system of shear wave imaging through high-frequency ultrasound elastography (HFUSE) to map the stiffness of mouse brains at different ages to verify how this process affected the mechanical properties of the organ. Both phantoms of PVA-C and gelatin were used in conjunction with a mechanical testing system to determine accuracy and resolution during HFUSE. Afterward, the proposed technique was also applied to *ex vivo* mouse brains. Through such experiments, it was possible to conclude that there was an error of

approximately 10 % in the elastic modulus measurements, and its axial resolution was 248 μ m. Moreover, while no statistical difference was obtained in the cortex, there was significant softening of the mouse brain with age.

Yin *et al.* [44] and Du *et al.* [46] both opted to use the phantoms as a supplementary way of validating their proposed techniques, together with simulation testing. Yin *et al.* [44], who proposed the development of a physics-informed neural network-based SWE (SWENet) method, used gelatin phantoms together with finite element simulations to prove that the use of the developed technology led to higher accuracy when it came to the spatial distribution of elastic properties. Du *et al.* [46], on the other hand, used a commercial phantom to validate the application of adaptive algorithms to improve the lateral resolution and contrast of an ultrafast transcranial US imaging technique with diverging wave transmission. Taking into account both the phantom experiments and the simulations, the authors concluded that these algorithms still suffered from phase aberration due to the refraction and attenuation innate to the human temporal bone. However, they still proved themselves capable of significantly improving the transcranial image quality, especially when it came to the image contrast.

Due to the great disparity in the materials used and considering the limitations of the current phantoms, there is a considerable need to develop a greater number of studies focused on the production of better brain and skull phantoms. In addition, the application of US technology to the brain remains relatively limited.

3.4. Heart

3.4.1. Motivation

It has been shown that changes in myocardial stiffness are associated with heart failure and increased mortality risk. Thus, various methods have been suggested for the assessment of the mechanical properties of the heart, including those based on strain analysis and the propagation of myocardial stretch. However, there is yet to be a widely accepted non-invasive method available in clinical practice to quantitatively assess myocardial stiffness [47].

Nevertheless, USE has proven itself to be a promising tool since it allows the evaluation of stiffness variation throughout the cardiac cycle. However, the heart possesses some characteristics that make its assessment with SWE more challenging, in particular its various anisotropic tissues, encompassing tendinous and muscular tissues. These structures comprise fibers organized in a complex helical distribution, and so stiffness will vary depending on the angle on which it is measured [47,48]. These characteristics may have contributed to a lack of research into the production of phantoms that accurately mimic the heart. In fact, of the obtained articles (Table 4), only one envisages the mimicking of porcine hearts using agarose phantoms. Although this particular study by Mishra and Cleveland [49] did not use elastography techniques and

Table 4

Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for the heart in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Agarose (0.25–2.0 % w/v)	Liver, kidney, heart	Determine the appropriate agarose concentrations that result in mechanical properties that match three different porcine organs	— *	4.4	N/A	N/A	Mishra and Cleveland, 2024 [49]
Agar (1, 1.5 wt%) + n-propanol + graphite	Tissue-mimicking	Assess the performance of the technique	HFUSE	N/A	4 $\pm$ 1.5	N/A	Lee et al., 2021 [50]
Zerdine® (CIRS, Model 039)	Liver	Validate the technique	SWE	N/A	1.74 $\pm$ 0.51	N/A	Pedreira et al., 2022 [47]

*No specific elastography technique is used to validate results, although these are compared with “data from the ultrasound elastography literature”.

only compared the results with existing data from elastography literature, this article was selected for analysis due to its primary focus on developing a heart-mimicking phantom. A series of agarose phantoms (with concentration ranging from 0.25 to 2 wt%) were developed and submitted to rheological testing to determine the optimal agar concentration. The obtained results were similar to the obtained from porcine organ samples (heart, kidney, liver) using the same methods. The agarose concentration of 0.4 wt% approximately matched the shear storage modulus of porcine heart and provided the best-fit stress relaxation in response to a step strain of 0.1 %.

3.4.2. Technique development

Lee *et al.* [50], whose goal was to develop a high-frequency ultrasound (HFUS) mechanical wave elastography system equipped with a motorized rotation stage for precise imaging of directional mechanics, also opted to use agar phantoms (1 wt% and 1.50 wt%), but these were not meant to represent any specific organ. Based on the results from the aforementioned study, it was apparent that the findings obtained using these phantoms may not accurately reflect what would occur in a real heart. Nevertheless, the authors decided to confirm the results by conducting tests using *ex vivo* porcine hearts, which allowed them to conclude that the developed elastography system was, indeed, capable of capturing the directional mechanics with high angular resolution.

In the study by Pedreira *et al.* [47], not only was there no artificial model created to replicate this organ, but the commercial phantom selected was designed to simulate liver fibrosis. These decisions led the authors to have to supplement their study with the use of specific *ex vivo* tissue samples and *in vivo* testing on healthy volunteers. The smart US approach successfully assessed the elastic properties of locally transverse isotropic soft tissues quantitatively, without the need for prior fiber orientation information, which is particularly valuable for organs like the heart, where fiber orientation can vary significantly.

Thus, it is noteworthy to mention that there is still a need to develop materials accurate enough to minimize the reliance on *ex vivo* tissues to validate results. It would also be interesting to explore phantoms capable of mimicking the particularities of the heart, such as its movement and the existence of different layers of tissue with different properties.

3.5. Muscles and tendons

3.5.1. Motivation

During the healing process, the reestablishment of fibers between the tendon and its surrounding tissues results in a variation of stiffness. In fact, several studies have indicated a correlation between lower tendon stiffness during recovery and poor functional outcomes. Conversely, regions with high SWV (increased stiffness) in injured tendons may indicate poor rehabilitation, potentially due to adhesion formation. For this reason, it is clear that knowledge of tendon stiffness may help clinicians make appropriate therapeutic strategies [51–54]. Besides tendons, it is also relevant to better understand muscle tissue stiffness. Changes in mechanical properties of skeletal muscle tissue are associated with many diseases, including spasticity, Duchenne's muscular dystrophy, hyperthyroidism, atrophy, paraplegia, patellofemoral syndrome, and myositis [55–57].

It is important to underscore that both muscles and tendons are inherently anisotropic, which might affect the accuracy of the measurements taken by SWE. Indeed, numerous studies have shown that the SWV in muscles and tendons is influenced by their fiber structure. Therefore, if fiber orientation is neglected, SWV measurements in muscles and tendons may not fully reflect their mechanical properties [52,53,55,56,57,58,59].

3.5.2. Material development

Muscle-mimicking phantoms that exhibit an anisotropic structure with known and adjustable mechanical parameters are warranted for validating the techniques that may be used in clinical settings. However, such types of phantoms are still lacking [56]. Table 5 summarizes the studies that have investigated the phantoms used in elastography in the last 5 years.

Two studies focused on the development of a more accurate phantom through the use of different fabrication techniques. Given the importance of heterogeneity and anisotropy, Guidetti *et al.* [57] proposed developing and characterizing a heterogeneous composite phantom with controllable properties, aiming to mimic the frequency-dependent anisotropic characteristics of skeletal muscle. To achieve that, 3D printing was the selected technique to fabricate the designed phantoms. It is relevant to point out that the printing process used in this study limits the achievable geometry for a muscle-mimicking material, preventing the creation of purely unidirectional fibers. As a result, the produced phantoms incorporated an additional layer of fibers for structural support. This material was formed by printing gelatin with a higher crosslinker concentration (15 % w/v) embedded in a softer matrix of the same material (5 % w/v). The composite phantoms exhibited anisotropic, heterogeneous, and viscoelastic properties. The results of the testing were compared to literature data from skeletal muscle tissue MRE experiments, which means that it isn't guaranteed that these phantoms would be as good when it comes to US techniques.

To overcome the limitations associated with regular 3D printing, Dong *et al.* [56] opted to modify the FRESH method through the increase of its printing size and the addition of a customized software for easy generation of the printing files, optimizing its original protocol for SWE. Moreover, acoustic scatterers were added in both the printing ink (sodium alginate) and the supporting bath, and hydrogel stiffness was varied by changing the concentration of crosslinking agent (Ca^{2+}). Through the use of this technique, the authors intended to achieve an alginate phantom that was capable of mimicking not only the macroscopic nature of the muscle but also its hierarchical microstructure at the cellular level, which, they claimed, could be simplified as a composite structure consisting of myofibers and extracellular matrix. Using the finite element simulations and the 3D-printed hydrogel phantoms, the proposed theoretical model for the estimation of skeletal-muscle-like composites was assessed. A good quantitative agreement between estimates and the values measured by SWE was achieved. To encompass the phantom anisotropy, SWE was performed in multiple directions ($\theta = 0^\circ: 10^\circ: 90^\circ$) by rotating the USE probe from the fiber direction to the cross-fiber direction.

An alternative approach was the development of anisotropic hydrogels, such as the one presented in [59]. In this study, Bied *et al.* proposed to test the acoustoelasticity theory for quasi-incompressible transverse isotropic media. This type of material is more accurate in the representation of muscle geometry. As such, transverse isotropic PVA phantoms were developed, and their results were complemented with the ones obtained for *ex vivo* beef muscle tissues. As expected, a linear relationship between the apparent shear modulus and stress was observed, and the slopes of the experimental curves enabled the determination of the nonlinear elastic coefficient of the transverse isotropic media under study. However, it was also verified that any deviation of the experimental positions from the theoretical configurations results in estimation errors of the nonlinear elastic coefficient.

These studies were particularly relevant due to their approaches towards manifestly anisotropic organs. However, a reliance on commercially available phantoms that do not accurately represent the intended study target persists. Ngo *et al.* [55], for example, recognized that it is not only important to take into account the anisotropic nature of skeletal muscles, but also the fact that they are organized into distinct

layers and anisotropic characteristics exist across various scales. They still opted to use homogeneous isotropic PVA phantoms to validate the proposed novel ultrafast US sequence consisting of steered pushing beams. They later supplemented the obtained results with testing on *ex vivo* fusiform beef muscles. In a final step, the authors also opted to apply the developed technique *in vivo* on fusiform muscles (*biceps braachi*) and mono-penate muscles (*gastrocnemius medialis*) during stretching and contraction. With this, it was possible to conclude that the developed technique allowed for the quantification of the shear modulus as well as the anisotropy factor (ratio of Young's moduli times shear moduli perpendicular to fibers) of the muscle using a single and fast acquisition.

Finally, Fernandes *et al.* [60] presented an innovative anisotropic ultrasound phantom composed of a bio-elastomeric silicone matrix reinforced with natural fibers, achieving sustainable and cost-effective mimicking of fibrous soft tissues such as skin and skeletal muscle. The phantoms exhibit tunable mechanical properties: Young's moduli ranged from approximately 7.16 to 11.04 MPa in skin-mimicking samples and 0.494 to 1.743 MPa in muscle-like constructs. Shear wave speeds varied between 1.5 and 5.93 m/s, while longitudinal wave speeds were in the 1086–1127 m/s range. The acoustic absorption coefficients ranged from 0.13 to 0.76 dB/cm/MHz, and the measured

storage, loss, and complex moduli fell within 1.035–6.652 kPa, 0.1831–0.8546 kPa, and 2.138–10.82 kPa, respectively. These results indicate that the phantoms reproduce anisotropic mechanical and acoustic responses akin to biological tissues. The material's directional stiffness and damping can be aligned with fiber orientation, making this system a promising platform for elastography, wave-propagation validation, and dynamic calibration in both clinical and research ultrasound applications.

3.5.3. Technique development

In another such case, Tsuchida *et al.* [61] developed a novel estimation method for vibration-based SWE measurements of human skeletal muscle. To evaluate the reliability of the proposed method, SWV was measured in both konjac and agar phantom gels, as well as skeletal muscle. Satisfactory intra-day, day-to-day, and inter-operator reliabilities were observed for both phantom and muscle SWV measurements, ensuring adequate reproducibility and reliability of the proposed estimation method.

Chen *et al.* [58] recognized the importance of taking into consideration the anisotropy of the muscle when using SWE so much so that they proposed the development of a customized, wearable T-shaped US

Table 5

Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for muscles and tendons in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Fibers: Gelatin (15 % w/v) Matrix: Gelatin (5 % w/v)	Skeletal Muscle	Have uniform controllable anisotropic properties and be comparable to skeletal muscle	MRE	N/A	1.73 ± 0.12	N/A	Guidetti <i>et al.</i> , 2019 [57]
Printing ink: sodium alginate sodium (2 wt %) + silicon dioxide (2 wt%) Supporting bath: gelatin (6 wt%) + calcium chloride (30 % v/v) + agar (2 wt%)	Skeletal Muscle	Validate the proposed theoretical framework	SWE	N/A	1.39 ± 0.24	N/A	Dong <i>et al.</i> , 2023 [56]
PVA (10 wt%) + cellulose (1 wt%)	Tissue-mimicking (anisotropic phantoms to more adequately represent the geometry found in the muscles)	Validate the theoretical development	SWE	N/A	N/A	N/A	Bied <i>et al.</i> , 2020 [59]
PVA (10 wt%) + cellulose (2 wt%)	Tissue-mimicking	Validate the proposed steered US pushing beam	SWE with custom-made algorithm developed as for the commercial sequence from Loupas and Defieux	N/A	2.76 ± 0.26	N/A	Ngo <i>et al.</i> , 2024 [55]
Konjac (commercially available) Agar (1–8 % w/v) + sterilized milk	Tissue-mimicking	Determine the reproducibility and reliability of the technique	SWE	N/A	2.21 ± 0.03	N/A	Tsuchida <i>et al.</i> , 2020 [61]
Zerdine ® (CIRS, Model 040 GSE)	Tissue-mimicking	Verify the performance and accuracy of optical tracking	DDSWE	20	2.58	N/A	Chen <i>et al.</i> , 2024 [58]
Zerdine ® (CIRS, Model 049)	Tissue-mimicking	Assess the technical feasibility of the technique	3D SWE	N/A	2.59 ± 0.03	N/A	Götschi <i>et al.</i> , 2021 [54]
Gelatin (15 wt%) + cellulose	Tissue-mimicking	Verify the performance of the technique	HFUSDDSWI	N/A	3.48 ± 0.06	N/A	Xu <i>et al.</i> , 2023 [53]
Gelatin (7–15 % w/v) + cellulose	Tissue-mimicking	Verify the performance of the technique	HFUSE with time-of-flight based method	N/A	4.02 ± 0.77	N/A	Chen <i>et al.</i> , 2021 [52]
Gelatin (7–15 % w/v) + cellulose	Tissue-mimicking	Verify the performance of the technique	HFUSE	N/A	2.585 ± 0.785	N/A	Chen <i>et al.</i> , 2020 [51]
Silicone (1A:1B) + Fibers (0.5 wt)	Tissue-mimicking (anisotropic phantoms to more adequately represent the geometry found in the muscles)	Have uniform controllable anisotropic properties and be comparable to skeletal muscle	SWE	111.85 ± 62.45	3.72 ± 2.21	0.445 ± 0.315	Fernandes <i>et al.</i> , 2025 [60]

transducer for dual-direction shear wave elastography (DDSWE) of the shoulder. The performance of DDSWE and the accuracy of optical tracking were verified by phantom experiments on commercial phantoms that did not accurately represent the human muscle. In consequence, the results had to be supplemented with human studies. Experimental results demonstrated that the proposed wearable US DDSWE exhibited a bias of approximately 6 % and a precision of 1.2 % for both directions in elastic phantoms. All the experimental results indicated that the proposed wearable US DDSWE could dynamically quantify the mechanical properties of shoulder muscles, aiding surgeons and physical therapists in adjusting the intensity of rehabilitation treatments.

The dependence on imprecise phantom models was particularly apparent in the context of tendon research. Indeed, none of the investigated studies utilized anisotropic phantoms, which would have provided a more realistic representation of tendon tissue.

Götschi *et al.* [54] proposed a method to create a volumetric reconstruction of mechanoacoustic properties of tendons by optically tracking the US probe during free-hand scanning. The study evaluated operational and technical feasibility, along with potential artifacts associated with hand-held image acquisition. This was performed using a synthetic phantom containing discrete targets with known mechanical properties. Again, since the phantom does not correctly translate the unique properties of the tissue under study, it was imperative to evaluate the technique reproducibility through inter-day and inter-reader scans of the patellar, Achilles, and supraspinatus tendons in ten healthy volunteers. The proposed approach demonstrated technical feasibility and reproducibility, with moderate to very good reliability and a standard error of measurement ranging from 0.3 to 0.6 m s⁻¹. This suggests that the proposed method may offer advantages over traditional two-dimensional measurements in terms of reproducibility and could be valuable for objectively assessing pathological tendon changes.

Xu *et al.* [53], similarly to Chen *et al.* [58], proposed the use of dual-direction shear wave imaging (DDSWI) to take into account the innately anisotropic nature of tendons. In this study, the authors decided to implement an HFUS shear wave imaging system based on two external vibrators. A phantom experiment using a generic phantom was then conducted to verify the performance of the proposed HFUS DDSWI technique, which had to be supplemented with human studies, where volunteers assumed three hand postures: relaxed, full fist, and tabletop. The experimental results revealed that flexion of the analyzed tendon led to an increase in both transverse and longitudinal SWVs. The authors concluded that the proposed HFUS DDSWI technique was a high-resolution imaging technique capable of characterizing the anisotropic properties of tendons in clinical applications.

Finally, two studies by Chen *et al.* [51,52] focused on the study of hand tendons. In these studies, an HFUSE system (handheld vibration system that was coaxially mounted with an external vibrator on an HFUS array transducer) was used to map the SWV of hand flexor tendons. In [51], through the use of gelatin phantoms, it was possible to observe that the biases of HFUSE were less than 6 % compared to standard mechanical testing approaches. Human experiments were also performed, showing the ability to use HFUSE to differentiate between the SWV of various hand tendons. On the other hand, in [52] phantom studies were used to quantify the performance of HFUSE, namely by measuring six parameters: bias, precision, lateral resolution, contrast, contrast-to-noise ratio, and accuracy. HFUSE exhibited excellent resolution (147 µm) in distinguishing the local stiffness of thin phantoms (1.2 mm) without compromising other performance metrics. Besides the phantom experiments, this study also relied on human studies, in which it was possible to map healthy and injured tendons. The developed system showed promise as a tool for evaluating the elastic properties of injured hand tendons post-surgery and throughout rehabilitation, offering a valuable way to monitor recovery progress.

Through this research, it was possible to verify that whilst the importance of taking into account the anisotropic nature of the muscle

and the tendon is often referred to in current studies, these were still heavily dependent on the use of isotropic phantoms, which forced them to supplement the obtained results with *ex/in vivo* testing. This being said, it was also possible to verify that the exploration of new fabrication techniques, such as 3D printing, might be a promising route to improve the mimetic nature of the phantoms.

3.6. Liver

3.6.1. Motivation

Chronic liver diseases represent a major public healthcare issue, affecting an estimated 844 million individuals worldwide, with a mortality rate of 2 million deaths annually [62]. There are several causes of liver disease, which include chronic hepatitis C virus, chronic hepatitis B virus, alcohol-related liver disease, and nonalcoholic fatty liver disease [63].

Chronic liver diseases follow a common pathophysiologic pathway in which repeated episodes of liver injury are followed by healing, regeneration, and fibrosis, ultimately culminating in an irreversible state called cirrhosis [64]. In essence, accurate diagnosis of the stage of liver fibrosis is crucial to track disease progression and assess treatment options. Elastography techniques provide a non-invasive method for the assessment of liver fibrosis by measuring liver stiffness, which increases in the presence of fibrosis. Hence, these techniques may enable early identification at a stage when reversibility is still possible, minimizing disease progression [65]. Table 6 summarises the studies that have investigated this technique in the last 5 years.

Given that elastography is currently being used in clinical practice for liver stiffness assessment, it is crucial to guarantee its accuracy and precision. To that end, to assess the influencing factors on the accuracy of liver elastography, a few studies have been conducted on commercial liver phantoms to analyze parameters associated with target properties such as stiffness and lesion size, in addition to acquisition parameters.

3.6.2. Technique development

Martiartu *et al.* [66] analyzed the sensitivity of SWV and shear wave dispersion (SWD) measurements to changes in acquisition parameters. The authors observed a greater variability of SWD, compared to SWV, mainly due to acquisition depth and lateral position of the ROI. The only significant source of variation for SWV was acquisition depth. Finally, although the variability caused by the ROI and the acquisition box sizes was insignificant, the authors suggested avoiding ROI diameters less than 10 mm to ensure proper measurement variability because of tissue heterogeneities.

Hwang *et al.* [67] observed that the inclusion's stiffness was increased when the inclusion was embedded in a stiffer background, with the authors suggesting that the difference between the stiffness of the inclusions and the surrounding material affected the transmission of shear waves at the boundary of inclusions. Furthermore, as observed in [66], the measurement depth impacted variability, indicating there could be a depth threshold to maintain the accuracy and reliability of the stiffness measurement.

In another phantom study using SWE [68], the stiffness measurements were also influenced by depth, particularly beyond 7 cm and inclusion size, with larger inclusions yielding more accurate measurements. The authors further evaluated the variability of the stiffness measurements *in vivo* on healthy livers for each hepatic segment. It was possible to assess stiffness within all hepatic segments despite the coefficient of variation being above 20 % in the deepest segments.

Racedo and Urban [69] investigated how varying the patch and window size affected the reconstruction of SWV in USE. It was found that large values for patch and window sizes yielded reliable measurements in commercial homogeneous phantoms with elasticity values meant to cover the range of healthy to cirrhotic-like liver stiffness. For the inclusion phantom, different trends were observed, with larger patch

Table 6

Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for the liver in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Zerdine ® (CIRS, Model 039)	Liver	Analyze the variability of SWV and SWD measurements	SWE	21.2 ± 20	3.57 ± 0.62	0.49 ± 0.02	Martiaru et al., 2021 [66]
Custom-made tissue-mimicking phantoms (CIRS) (Material isn't described)	Liver	Investigate the factors influencing measurements	SWE	16 ± 2	N/A	N/A	Hwang et al., 2019 [67]
Zerdine ® (CIRS, Model 039 and Model 049A)	Liver Tissue-mimicking	Assess the accuracy and variability of the technique	SWE	N/A	6.63	N/A	Moalla et al., 2020 [68]
Zerdine ® (CIRS, Model 039 and Model 049A)	Liver Tissue-mimicking	Explore the effects of the patch and window values on the SWV reconstructions	SWE	21.2 ± 20	2.4 ± 1.45	0.47 ± 0.02	Racedo and Urban, 2019 [69]
Elastic and viscoelastic phantoms (CIRS) (Material isn't described)	Liver	Quantify the bias of SWV measurements	SWE, TE, MRE	N/A	2.025 ± 0.051	N/A	Palmeri et al., 2021 [96]
Polyacrylamide (13 wt%) + glycerin (1 wt%) + aluminum oxide (3 wt%)	Liver	Quantify the bias of SWV measurements	SWE, TE, MRE	N/A	1.8 ± 0.4	0.5 ± 0.01	Kishimoto et al., 2022 [97]
Zerdine ® (CIRS, Model 039)	Liver	Assess repeatability and agreement of SWVs across 6 US SWE systems	SWE	N/A	2.2 ± 1.4	N/A	Gilligan et al., 2020 [98]
Zerdine ® (CIRS, Model 049)	Tissue-mimicking	Assess the feasibility of the technique	SWE	45 ± 35	N/A	N/A	Hwang et al., 2020 [99]
Custom-made tissue-mimicking phantoms (CIRS) (Material isn't described)	Liver	Compare the FibroTouch device with the established FibroScan	TE	3.1	N/A	N/A	Serra et al., 2020 [100]
Polyacrylamide (20 % w/v) + microcrystalline cellulose (0.25 % w/v)	Liver	Mimic SWV-dispersion of the human liver	SWE, Time harmonic elastography, MRE	N/A	0.08	0.37	Morr et al., 2022 [101]
Zerdine ® (CIRS, Model 039) Commercial elasticity phantoms KS215 T-3 (Institute of Acoustics, Chinese Academy of Science, Beijing, China) (Material isn't described)	Liver	Evaluate the accuracy, feasibility and validity of the technique	TE	40 ± 32	N/A	N/A	Huang et al., 2024 [70]
Gelatin (3–7 wt%+ cellulose (1 wt%))	Liver	Demonstrate the generation and propagation characteristics of focused shear wave beams	TE	N/A	2 ± 1	2.5	Cormack et al., 2024 [71]
Agar (3 wt%) + gelatin (5, 10 wt %)	Liver	Technique validation	Quasi-static elastography	40.81 ± 4.20	N/A	N/A	Selladurai and Thittai, 2019 [72]
Agar (3 wt%) + gelatin (5 wt%)	Liver	Technique evaluation	Quasi-static elastography	19 ± 4.90	N/A	N/A	Selladurai et al., 2020 [73]
Elastic and viscoelastic phantoms (CIRS) (Material isn't described)	Tissue-mimicking	Technique validation	Quasi-static elastography with STREAL method	1	N/A	N/A	Aboutaleb et al., 2022 [74]
Zerdine ® (CIRS, Model 039)	Liver	Technique validation	External mechanical vibration-based 3D SWE	N/A	N/A	N/A	Huang et al., 2020 [76]
Zerdine ® (CIRS, Model 039)	Liver	Technique validation	S-WAVE	3 ± 12	N/A	N/A	Zeng et al., 2021 [77]
Zerdine ® (CIRS, Model 039 and Model 049A)	Liver	Technique validation	S-WAVE	N/A	N/A	N/A	Hashemi et al., 2023 [78]
Gelatin (7.5 % w/v) + sodium dodecyl sulfate (0.5 % w/v) + castor oil (1 wt%)	Tissue-mimicking	Technique validation	Torsional Wave Elastography	N/A	1.75 ± 0.5	N/A	Faris et al., 2020 [79]
Gelatin (7.5 % w/v) + sodium dodecyl sulfate (0.5 % w/v) + graphite particles (0.5 % w/v)	Tissue-mimicking	Technique validation	Torsional Wave Elastography	N/A	1.75 ± 0.5	N/A	Faris et al., 2020 [79]
Gelatin (11.1 % w/v) + evaporated milk (50 % v/v)	Tissue-mimicking	Technique evaluation	MR-SWE	N/A	2.52 ± 0.18	N/A	Hofstetter et al., 2019 [80]
Custom-made tissue-mimicking phantoms (CIRS) (Material isn't described)	Tissue-mimicking	Investigate the effects of various artifacts on measurements and validate the effectiveness of the technique	Vibration-guided transient elastography	N/A	N/A	N/A	Lorée et al., 2020 [81]

(continued on next page)

Table 6 (continued)

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Zerdine ® (CIRS, Model 039) Gelatin	Tissue-mimicking	Technique validation	Robotic SWE System	15 ± 3.1	1.99 ± 1.58	N/A	Czernusiewicz et al., 2022 [82]
Zerdine ® (CIRS, Model 06GSE)	Tissue-mimicking	Technique validation	STL-SWE	N/A	2.73 ± 0.34	N/A	Ahmed et al., 2020 [83]
Custom-made tissue-mimicking phantoms (CIRS) (Material isn't described)	Tissue-mimicking	Test the robustness of the method	SWE with the 2P-CWT method	N/A	4.5 ± 2.5	0.45	Kijanka et al., 2019 [86]
Custom-made tissue-mimicking phantoms (CIRS) (Material isn't described)	Tissue-mimicking	Examine the reliability of the method used for shear wave phase velocity curves evaluation	SWE with the GST-SFK method	N/A	N/A	0.5	Kijanka and Urban, 2021 [87]
Custom-made tissue-mimicking phantoms (CIRS) (Material isn't described)	Tissue-mimicking	Explore how the phase velocities in inclusions varies with frequency	SWE with the LPVI method	N/A	4.69	0.5	Kijanka and Urban, 2021 [88]
Zerdine ® (CIRS, Model 039 and Model 049A) A gelatin phantom with an excised porcine liver inclusion	Tissue-mimicking	Test the robustness of the method	SWE with the LPVI method	35 ± 10	3.55 ± 0.43	0.5	Kijanka and Urban, 2019 [89]
Block of konjac Homogeneous soft tissue phantom (OST. Co. Ltd., Japan)	Tissue-mimicking	Evaluate viscoelasticity using C-SWE with multiple frequency selections	C-SWE	N/A	7.20 ± 1.79	N/A	Koda et al., 2024 [90]
Gelatin (5 wt%) + cellulose (1.5 wt%) + xanthan gum (0.1–0.6 wt%) Inclusion: Gelatin (5 wt%) + cellulose (1.5 wt%) + xanthan gum (0.2, 0.5 wt%) + graphite (0.1 wt%)	Tissue-mimicking	Investigate the feasibility of viscosity reconstruction maps	SWE with Frequency-Shift Method	N/A	N/A	N/A	Bhatt et al., 2019 [91]
Custom-made tissue-mimicking phantoms (CIRS) (Material isn't described)	Tissue-mimicking	Examine the robustness of the 2P-FSP approach for the shear wave attenuation measurement	SWE with the 2P-FSP method	N/A	N/A	0.5	Kijanka and Urban et al., 2022 [92]
Gelatin (5 wt%) + dietary xanthan gum (0.1, 0.5 wt%) + cellulose particles (1.5 wt%)	Tissue-mimicking	Validate and compare the performance of different US shear wave attenuation imaging techniques	SWE with the R-FS method	9.35 ± 0.35	3.32 ± 0.18	N/A	Yazdani et al., 2022 [93]
Gelatin (1 % w/v) + dawn soap (5 % w/v) + castor oil (5–20 wt %) + corn starch (1 % w/v)	Tissue-mimicking	Technique testing	SWE with the pSTL-FS method	N/A	N/A	N/A	Mislati et al., 2024 [94]
Zerdine ® (CIRS, Model 039) Gelatin (3, 6 wt%) + oil (40 wt %) Gelatin (3, 6 wt%)	Tissue-mimicking	Technique validation	SWE STL-TRSS	N/A	N/A	N/A	Khan et al., 2024 [95]
PVA-cryogel	Tissue-mimicking	Test the robustness of the method	US	8.85 ± 7.25	N/A	N/A	Cournane et al., 2012 [102]
PVA-cryogel	Tissue-mimicking	Test the robustness of the method	US	8.85 ± 7.25	N/A	0.7 ± 0.3	Cournane et al., 2020 [27]

and window sizes improving contrast-to-noise ratio but smaller sizes reducing bias.

Huang et al. [70] developed a new, wireless, palm-sized TE system that enables liver stiffness measurement guided by real-time B-mode imaging. The system was validated in multiple commercial elasticity phantoms and in adults with chronic liver diseases, showing sufficient accuracy and high agreement of liver stiffness measurements with conventional TE.

Cormack et al. [71] proposed a method to increase shear wave signal at the required depths in liver stiffness assessment by focusing the waves into a directional beam. Gelatin phantoms (3 wt% and 7 wt%), containing 1 wt% cellulose powder were created to measure SWV in a media with shear stiffnesses covering the range seen in healthy and fibrotic liver tissues. It was demonstrated that focused shear wave beams had greater amplitudes in the liver stiffness measurement region compared to unfocused shear waves, suggesting an increase in the accuracy of measurements for US TE.

Two studies were conducted to optimize quasi-static USE for liver imaging. Selladurai and Thittai [72], to quantitatively image the

unknown modulus of the phantom, interposed a calibrated reference layer with known stiffness between the transducer and the liver-mimicking phantom, made with agar and gelatin. The results demonstrated the possibility of estimating the compressive elastic modulus of the phantom using quasi-static USE and a calibrated reference material within an error of 11 %. The authors further investigated this approach on *ex vivo* goat liver samples [73]. A recent work [74] proposed a method to reconstruct the elastic modulus for quasi-static USE, using an initial displacement estimate obtained by conventional time delay estimation methods and a strain refinement algorithm (STREAL). High-quality strain and elasticity images were obtained for both the commercial tissue-mimicking phantom and a liver cancer patient.

More recently, 3D SWE has caught the attention of researchers, as it captures the volumetric shear wave pattern of the liver tissue, allowing more accurate SWV estimation and better quantification of liver stiffness [75]. Huang et al. [76] developed an external mechanical vibration-based 3D SWE technique that can be integrated into a clinical US scanner, able to measure SWV at low volume rates and with a penetration depth of up to 10 cm, exceeding the capacity of the ARF-based

methods. Additionally, Zeng *et al.* [77] presented a novel US matrix array implementation called shear wave absolute vibro-elastography (S-WAVE) that uses 3D imaging to estimate tissue elasticity over a tissue volume. This technique showed a strong agreement with the results obtained from MRE, however, the acquisition time was 12 s, which required patients to hold their breath for at least that long. Hence, to make the imaging faster, they further combined ultrafast elastography with the S-WAVE system [78]. This method accurately predicted tissue elasticity, showing less than 9 % error compared to MRE, and reduced the acquisition time to 0.05 s. All these studies used commercial phantoms from CIRS.

Although tissue-mimicking phantoms remain the first choice to validate new techniques, some studies have utilized *ex vivo* or *in vivo* tissue. Faris *et al.* [79] validated the Torsional Wave Elastography technique against SWE by comparing their measurements of SWV, shear moduli, and viscosity of *ex vivo* chicken liver samples and tissue-mimicking hydrogel phantoms. The results of shear moduli for *ex vivo* liver showed a good agreement between the two techniques (1.69–4.0 kPa using Torsional Wave Elastography and 1.32–4.48 kPa using SWE) for a range of frequencies from 200 to 800 Hz, aligning with other results obtained from the literature. Hofstetter *et al.* [80] presented a novel MR-SWE method that measures the speed of propagating wave packets generated by ARFI. The acquisition and combination of multiple-point data resulted in high-resolution maps of SWV. This method detected a significant increase in SWV from $1.65 \pm 0.18 \text{ m s}^{-1}$ in normal *ex vivo* liver tissue to $2.52 \pm 0.18 \text{ m s}^{-1}$ in the ablated region following focused US thermal ablation treatment. Lastly, Lorée *et al.* [81] introduced a novel method called vibration-guided transient elastography to facilitate the localization of an optimal measurement window. By accurately detecting artifacts such as blood vessels and arteries, which can interfere with shear wave propagation, the operator could rapidly locate optimal measurement windows and improve the accuracy of stiffness measurements of *in vivo* liver tissue.

Two other studies used small animal models to validate new elastography techniques for preclinical research. Czernuszcwicz *et al.* [82] developed a robotically controlled SWE imaging platform for assessing liver stiffness and echogenicity in mouse models of liver disease. This platform demonstrated its ability to detect small changes in liver disease state, enabling *in vivo* staging of rodent liver disease. Ahmed *et al.* [83] developed a software-based automated respiration gating technique to adapt Single Track Location SWE (STL-SWE) for imaging mouse models, mitigating the effects of respiratory motion on the assessment of mechanical properties. Using a murine model of pancreatic cancer liver metastasis, they detected stiffness changes occurring over long time scales (months).

Most of the referenced articles use SWE systems that assume the medium to be elastic and linear, employing time-of-flight methods to monitor the propagation of shear waves over time and correlate their spatiotemporal coordinates to determine SWV [84]. However, modeling soft tissue that exhibits viscoelasticity as purely elastic limits the accuracy of elasticity measurements. In a viscoelastic medium, the SWV varies with frequency, a phenomenon called dispersion. Tissue viscoelasticity is often studied by analyzing shear wave phase velocity curves, but it can also be characterized by measuring the shear wave attenuation. Two common methods to measure phase velocities include the phase gradient and 2D-FT [85]. Recently, alternative approaches have been developed and tested mainly using tissue-mimicking phantoms and *ex/in vivo* liver.

Kijanka *et al.* [86] presented a novel two-point continuous wavelet transform (2P-CWT) method to estimate shear wave phase velocity dispersion based on shear wave propagation measurements at only two lateral points. In *ex vivo* porcine the 2P-CWT method, compared to the 2D-FT technique, showed a smoother trend for phase velocity up to 600 Hz. Moreover, the 2D-FT approach required shear waves to travel a longer lateral distance than the 2P-CWT method to stabilize the mean phase velocity. Another technique called GST-SFK [87], outperformed

the 2D-FT, yielding robust dispersion curves over a broader frequency range, as high as 800 Hz and 700 Hz for *ex vivo* and *in vivo* human livers of six healthy volunteers, respectively. The usable bandwidth was at least doubled compared to the other methods. Kijanka and Urban [88] proposed the local phase velocity-based imaging (LPVI) method to create images of phase velocities and viscoelastic parameters. In contrast to their previous work where they first developed this technique [89], bandpass filters were added to increase the frequency range for which 2D phase velocity maps could be reconstructed. Using commercial tissue-mimicking viscoelastic phantoms and *in vivo* liver data, this method showed good reconstructions of 2D shear wave phase velocity maps. More recently, a continuous shear wave elastography (C-SWE) method that allows arbitrary frequency selection of the shear wave to measure liver viscoelasticity by combining measurements at different frequencies was introduced [90]. The SWV dispersion measured for four healthy liver volunteers ranged from 5.41 and $8.98 \text{ m s}^{-1} \text{ kHz}^{-1}$, comparable to the values reported using MRE.

To compute shear wave attenuation, Bhatt *et al.* [91] proposed and tested the frequency shift method (FS). Homogeneous and heterogeneous tissue-mimicking phantoms were prepared using 5 wt% gelatin, 1.5 wt% cellulose particles acting as scatterers, and various concentrations of xanthan gum (0.1–0.6 wt%) to model viscosity. Inclusions contained 0.1 wt% graphite powder to mimic the appearance of lesions. Successful viscosity mapping was achieved in both homogeneous and heterogeneous phantoms. Analysis of animal tissues showed that the *ex vivo* fatty duck liver and *in vivo* fatty goose liver samples were more viscous than the healthy *ex vivo* porcine liver samples. Kijanka and Urban [92] further modified the aforementioned method. This method, called 2P-FS, uses the frequency shift of the power spectra for the shear wave motion data measured at only two spatial locations. The results showed a proper reconstruction of the shear wave attenuation for a further lateral location counting from the push beam location, compared to 2P-FS in commercial tissue-mimicking viscoelastic phantoms. However, it did not demonstrate significant improvement in *ex vivo* porcine livers due to limited lateral locations available beyond the push beam location. Yazdani *et al.* [93] proposed a revisited frequency shift (R-FS) method to improve the robustness of shear wave attenuation imaging for medical diagnosis of steatotic livers. This method assumes a variable shape parameter with distance and employs an adaptive random sample consensus line fitting method. Compared to FS and 2P-FS, the R-FS method produced attenuation maps with lower bias and variance *in vivo* duck liver datasets and provided estimates closer to attenuation measuring US shear wave elastography (AMUSE). Mislati *et al.* [94] extended the pSTL-SWE technique by incorporating the FS approach (pSTL-FS) and evaluated its effectiveness in reconstructing attenuation coefficient maps by fitting the shear wave spectrum to a Gamma or Gaussian distribution. In viscoelastic phantoms made from 10 % w/v gelatin and various concentrations of castor oil (5, 16, and 20 wt%), assuming a Gamma distribution for pSTL-FS resulted in higher accuracies ranging from 84.69 % to 97.55 %. The attenuation coefficient estimates from *in vivo* murine liver revealed that the pSTL-FS (Gamma) approach exhibited the most significant difference between control and obese mice, compared to other methods.

Finally, Khan *et al.* [95] developed a STL Time Resolved Shear Wave Spectroscopy (STL-TRSS) approach that captures both the attenuation and dispersion of shear waves to estimate viscoelastic stiffness. The results showed that when the tissue exhibits viscoelastic properties, the complex modulus is less prone to error in estimating tissue stiffness than the elastic shear modulus, estimated by applying a cross-correlation estimator.

3.6.3. Equipment reliability

Moreover, another aspect that should be considered is the consistency in measurements across different elastography systems.

In this regard, as a way to quantify the bias of SWV measurements between different commercial ultrasonic shear elasticity systems and an

MRE system, the Quantitative Imaging Biomarkers Alliance (QIBA) performed a phantom study, using elastic and viscoelastic commercial phantoms that mimic liver stiffness [96]. The results indicated a more consistent SWV reconstruction across different manufacturer systems in elastic than in viscoelastic phantoms. Statistically significant differences in SWV measurements were identified among different systems, sites, and focal depths. Another study [97] investigated the bias of SWV measurements across several USE systems using a viscoelastic phantom and compared the results with those from TE and MRE. The phantom with stiffness representing fibrotic liver consisted of 13 wt% acrylamide, 1 wt% glycerin, and 3 wt% aluminum oxide powder. SWV measurements were influenced by the probe type, and although measurement biases associated with the SWE systems were observed, they were not necessarily consistent. The focal depth was a less significant source of SWV variability than the system. Moreover, the SWV of the viscoelastic phantom obtained using different techniques increased according to the shear wave frequency used, with the SWV measured by the SWE systems being significantly higher than the other two. Other studies opted to study the performance of elastography using phantoms along with human livers. Gilligan *et al.* [98] assessed the intersystem agreement and repeatability of SWV measurements across 6 US SWE systems. The study conducted on commercial elastic phantoms and *in vivo* human livers, demonstrated an overall agreement among systems. Hwang *et al.* [99] evaluated the SWE technique of ElastQ imaging and compared it with SuperSonic shear imaging using a commercial tissue-mimicking phantom and *in vivo* liver from patients with various liver diseases. Both systems demonstrated high repeatability in elasticity measurements of phantom targets and excellent inter-observer agreement. The *in vivo* study revealed a strong correlation between stiffness measurements obtained using ElastQ and SuperSonic shear imaging. In turn, Serra *et al.* [100] compared the performance of the established FibroScan device with the new FibroTouch device in assessing liver stiffness, both in commercial liver-mimicking phantoms and patients. They reported a good correlation between FibroScan and FibroTouch for stiffness, though FibroTouch had a larger variation and tended to overestimate liver stiffness.

To standardize liver elastography across different devices, Morr *et al.* [101] developed and characterized a liquid-liver phantom that mimics the SWV dispersion of a healthy human liver and can be produced reproducibly. The phantom was made of linear polymerized polyacrylamide and microcrystalline cellulose (0.5 μm) was added to provide US speckle contrast. It was observed that the phantom reproduced the viscoelastic dispersion of healthy human liver, standing out as an alternative to commercial phantoms that are purely elastic. Additionally, the shear-wave damping was slightly higher compared with healthy human liver *in vivo*, but they suggested that a mixture of crosslinked and linearly polarized polyacrylamide chains may better mimic the damping of the human liver. Finally, the properties did not change considerably with temperature and remained stable from weeks 1–2 of curing.

The interaction between phantom design and imaging accuracy, particularly in pathologies or complex tissue overlays, merits close scrutiny. Cournane *et al.* (2012) [102,103] demonstrated that fatty deposits overlaying liver-mimicking phantoms introduced significant bias in transient elastography measurements. Using PVA-C phantoms with varied fat-like layers, they found that fat layers thicker than ~ 45 mm led to overestimations of stiffness by up to 11 kPa, and simulated steatosis conditions produced approximately 54 % overestimation by FibroScan® systems. This highlights that mismatches in acoustic velocity and assumptions embedded in commercial elastography devices can severely distort diagnostic outputs when encountering realistic anatomical conditions. Complementing this, Cournane *et al.* (2020) [27] addressed quality metrics in liver shear-wave mapping, demonstrating that implementing a quality threshold algorithm reduced the measurement variability induced by respiratory motion, especially in paediatric subjects undergoing free-breathing scans. This algorithm improved differentiation of fibrosis grades by suppressing low-quality data points,

thereby enhancing diagnostic reliability. Collectively, these works underscore that phantom realism must not only focus on tuneable mechanical and acoustic properties but also consider layered anatomical realism and motion-induced artifacts. Without this comprehensive fidelity, elastography outcomes risk both clinical misclassification and regulatory scepticism.

Over the past 5 years, liver elastography has emerged as a significant research interest, with a particular focus on developing innovative techniques. As mentioned, the liver is an organ that exhibits viscoelastic behaviour, and although some researchers manifested interest in studying and mimicking this behaviour, most of the analyzed studies used phantoms that only represented the elastic component of the liver, with a vast majority relying on commercial phantoms. As is clear, there is a growing need to use phantoms that accurately represent liver tissue to validate these techniques, enabling their widespread use in clinical practice, reducing the reliance on *in vivo* and *ex vivo* studies.

3.7. Breast

3.7.1. Motivation

Breast cancer has overtaken lung cancer as the leading cause of global cancer incidence, with an estimated 2.3 million new cases in 2020, accounting for 1 in 8 cancer diagnoses. It is the fifth leading cause of cancer mortality worldwide, with an estimated 685,000 deaths [104]. If current trends persist, the global burden of breast cancer is expected to increase by over 40 % to approximately 3 million new cases annually by 2040 [105].

Mammography and ultrasonography are the current standard clinical imaging modalities for breast cancer screening, but both have limitations. Mammography is less effective for dense breasts, less sensitive to small tumors, and does not provide any indication of eventual disease outcome [106], while ultrasonography has low specificity, leading to a high false-positive rate and unnecessary recalls and biopsies [107]. USE can be used for the preliminary assessment of malignancy, potentially reducing the need for follow-up biopsies.

Although breast elastography is not widely available as a standard imaging option in healthcare facilities, ongoing research is focused on developing novel techniques and addressing current limitations (Table 7).

3.7.2. Material development

Regarding the development of breast phantoms, two studies stand out as significant contributions over the last 5 years. Leonov *et al.* [108] developed anatomical breast phantoms made from PVC plastisol with graphite powder and metallic glitter as scatterers. These phantoms were designed to practice US diagnostic skills, elastography imaging, and US-guided biopsy sampling. The stiffness of lesions and breast tissue in the phantoms was modeled using various concentrations of plasticizer, with higher concentrations resulting in stiffer materials. By varying scatterer concentration, lesions with different echogenicity were produced: hypoechoic simulating a fibroadenoma or a malignant tumor, and hyperechoic mimicking a lipoma. Additionally, the measured speed of sound and acoustic attenuation varied with the concentration of scatterers, matching the reported values for human breast tissues. Attenuation at 1 MHz and speed of sound showed an increase from 0.05 dB cm^{-1} MHz $^{-1}$ (no scatterers) to 0.45 dB cm^{-1} MHz $^{-1}$ and from 1408 m s $^{-1}$ to 1522 m s $^{-1}$, respectively, with the addition of scatterers. Lastly, echograms delineated lesion boundaries, demonstrating the effectiveness of the phantoms in simulating real breast tissue. Aldosary *et al.* [109] constructed a silicone-based breast phantom reproducing several components of the breast, appropriate for oncological breast surgery practice maneuvers, multi-modality imaging, and radiation therapy dose measurements. The phantom's effectiveness in simulating real breast tissue was validated through surgical simulations, imaging studies (computed tomography, MR imaging, and US), and dosimetric measurements. All imaging modalities displayed the various anatomical

Table 7

Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for breast in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Gelatin (4.5, 6.5 wt%) + agar (1.6–3 wt%)	Breast	Technique testing	Combination of quasi-static elastography and SWE	N/A	N/A	N/A	Wang <i>et al.</i> , 2022 [110]
Zerdine® (CIRS, Model 059)	Breast	Technique testing	Quantitative elastography from B-mode images	60.0 ± 4.0	2 ± 1	N/A	Rabin and Benceh, 2019 [111]
Zerdine® (CIRS, Model 059)	Breast	Evaluate image quality of elastograms in terms of contrast-to-noise ratio and non-uniformity level	Quasi-static elastography	20	N/A	N/A	Matrone <i>et al.</i> , 2019 [112]
Tissue mimicking phantom (CIRS) (Material isn't described)	Tissue-mimicking	Test the effectiveness of US tissue motion tracking techniques	Quasi-static elastography	60	N/A	N/A	Caius and Samani, 2023 [113]
Zerdine® (CIRS, Model 059)	Breast	System validation	SWE	17.5 ± 0.1	N/A	N/A	Kelly <i>et al.</i> , 2018 [114]
Gelatin (40.8 g) + oil (612 mL) + dishwashing liquid (12 mL) Inclusions: Agar (3 % w/v)	Breast	Technique testing	Quasi-static elastography	N/A	N/A	N/A	Di Meo <i>et al.</i> , 2020 [115]
Zerdine® (CIRS, Model 059)	Breast	Train and validate the model	Transfer Learning US Elastography	N/A	N/A	N/A	An <i>et al.</i> , 2023 [116]
PVC plastisol + graphite powder (0.5, 1 wt%) + metallic glitter (0.5 wt%)	Breast	Practice US diagnostic skills	Quasi-static elastography	N/A	N/A	0.225 ± 0.2	Leonov <i>et al.</i> , 2023 [108]
Silicone rubber (140 g) Silicone rubber (600 g) + silicone oil (300 g)	Breast	Provide a realistic model to mimic breast tissue for surgical simulation	SWE	13.4 ± 4.2	N/A	N/A	Aldosary <i>et al.</i> , 2020 [109]
Zerdine® (CIRS, Model 049)	Tissue-mimicking	Determine the optimal acoustic radiation push location	SDUV	18.75	5.59	N/A	Kumar <i>et al.</i> , 2018 [117]
Soft silicones	Breast	Verify the consistency of two elastography techniques	C-OCE, SE	260 ± 160	N/A	N/A	Gubarkova <i>et al.</i> , 2022 [118]

features, and the elasticity measurements obtained from the breast phantom using SWE and mechanical testing were consistent with those reported for fibroglandular breast tissue. Despite the overall agreement, variability in Young's modulus measurements between the two techniques was observed, aligning with the reported variability between different testing methods. Although not designed for elastography, this breast phantom can be used to evaluate the reliability and reproducibility of elastography measurements.

3.7.3. Technique development

Wang *et al.* [110] addressed the challenges of quantifying linear elastic properties at small deformation levels in USE, which may hinder the accurate differentiation between malignant and benign tumors. They combined quasi-static and SWE to non-invasively estimate the localized stress–strain curve of materials at large deformation. Agar-gelatin phantoms mimicking human breast tissue and fibroadenomas were used to evaluate the method. The stress–strain curves obtained were consistent with those from the unconfined mechanical test, accurately distinguishing materials with different elastic properties. Another study [111] attempted to overcome the limited clinical accessibility of elastography by developing a quantitative elastography method based on B-mode US images acquired by conventional US scanners. They digitized the US scanner's video output signal to generate elasticity maps from a sequence of B-mode images acquired during the propagation of a diffuse field within the medium. The elasticity values using B-mode images of the breast phantom and values reported by the phantom's manufacturer showed good agreement. Additionally, the average Young's modulus values aligned well with the reported values in the literature for breast carcinomas and normal fibroglandular tissue.

Two other studies focused on displacement estimation techniques

used in quasi-static elastography to reconstruct elastograms. Matrone *et al.* [112] compared the performance of a coherence-based beam-forming technique with the standard Delay and Sum (DAS) beamforming in improving the displacement and strain estimation in quasi-static elastography of the breast. Coherence-based methods enhanced lateral displacement accuracy but offered comparable performance for axial displacement estimation. Furthermore, the image quality of elastograms obtained from a commercial breast phantom did not surpass that of DAS. In turn, Caius and Samani [113] evaluated the performance of four different tissue displacement estimation methods that can be used in conjunction with STREAL to estimate tissue Young's modulus reconstruction level. The assessment was performed using datasets from US imaging of commercial tissue-mimicking phantoms, as well as clinical data. The authors found that the analytic minimization in 2D algorithm, especially when enhanced with STREAL, produced high-quality images for a very low runtime requirement.

An interest in multimodal approaches is also evident, with the contributions of Kelly *et al.* [114] and Di Meo *et al.* [115]. In the first study, the authors developed an automated breast US scanner that integrates SWE, Doppler flow imaging, and photoacoustic tomography. This system demonstrated its ability to accurately measure elasticity, detect flow, and effectively combine elasticity and photoacoustic data to provide high contrast between stiffer inclusions and a softer background in phantoms. Di Meo *et al.* [115] proposed a multimodal approach combining millimeter-wave imaging, US imaging, and elastography for breast cancer detection. A homogeneous oil-in-gelatin phantom was fabricated to mimic low-density breast tissues, and a water-agar inclusion with 3 % w/v agar was embedded to simulate a tumor. The millimeter-wave imaging system detected a target at a depth of approximately 1.4 cm as a high-intensity spot, demonstrating its ability

to identify non-surface targets in a phantom with significant dielectric losses. In the B-mode image, the target appeared as a hypoechoic mass and as a stiff inclusion in elastography images.

More recently, deep learning techniques have been investigated in medical image analysis. In [116] An *et al.* developed a transfer learning USE model to quantitatively classify breast masses as benign or malignant based on elastograms. The NewNet model, evaluated on a commercial breast phantom with spherical inclusions, achieved an accuracy of 91 % in classifying elastogram types but struggled to accurately predict inclusion sizes, with an accuracy of only 33 %. Compared to other transfer learning models, the NewNet model yielded equivalent results.

A recurrent use of breast phantoms to validate methods and techniques is noticeable. However, some studies opted for using *ex vivo* or *in vivo* breast tissue. Aiming to evaluate the viscoelastic parameters in breast tissue, Kumar *et al.* [117] conducted a study on a group of patients using shear wave dispersion ultrasound vibrometry (SDUV). Firstly, using a commercial tissue-mimicking phantom, the authors determined that the acoustic radiation push needed to be focused outside the suspicious mass boundary in the normal tissue region to limit the variance in acoustic characteristics of the shear wave. The *in vivo* measurements showed a significant difference in shear viscosity between malignant and benign tissues, malignant and normal tissues, and benign and normal tissues. Although shear viscosity varied significantly among malignant masses, suggesting that breast cancer pathologies might have different viscosities, relaxation time (defined as the ratio of shear viscosity to elasticity) had a smaller spread, making it a more promising biomarker. In another study [118], the performance of SE and compression OCE (C-OCE) in differentiating benign and malignant breast lesions *in vivo* was compared. Unlike SE, C-OCE with pre-calibrated reference layers enabled the quantitative estimation of local stress applied to the tissue during deformation. Consequently, C-OCE accurately classified all lesions, agreeing with histology, whereas SE misclassified some lesions.

Similar to the liver, breast tissue exhibits viscoelastic properties, and, as it can be noticed, most phantoms do not replicate this behavior, which may result in a biased estimation of elasticity. In that regard, to address that deficiency, some researchers used *ex/in vivo* breast tissue. In essence, further research is imperative to develop breast phantoms that accurately mimic tissue behavior and anatomy, thereby reducing the reliance on organ studies.

3.8. Prostate

3.8.1. Motivation

Prostate cancer is the second most frequent cancer diagnosed in men and the fifth leading cause of death worldwide, counting 1.2 million new cases and 350 thousand deaths in 2018.[119] The risk of prostate cancer is typically assessed using a combination of digital rectal examination and prostate-specific antigen testing. If the prostate-specific antigen level or digital rectal examination is abnormal, a histopathological examination is performed through transrectal ultrasound-guided systematic biopsy [120]. However, diagnosis is often missed in many patients with biopsies missing or underdiagnosed aggressive tumors. Furthermore, increasing the number of biopsy cores could lead to overdiagnosis by identifying clinically insignificant cancers [120,121].

To overcome the limited diagnostic accuracy of conventional biopsy methods and increase the cancer detection rate, advanced techniques capable of differentiating between neoplastic and normal prostate tissue lesions are currently being developed. Measurements of prostate tissue stiffness on the entire prostate or on biopsies from men with suspected prostate cancers revealed that malignant prostate tissues are almost 60 % stiffer than benign prostate tissues [122]. The increased stiffness is correlated with an increased alignment of collagen fibers as prostate cancer becomes more aggressive. In this regard, USE, by measuring tissue stiffness, emerges as a reliable tool to assist clinicians in detecting prostate cancer. Table 8 summarises the studies that have investigated this technique in the last 5 years.

Transrectal SWE is used to measure the elasticity of the prostate gland. To evaluate the variability of this technique, a study was conducted using a phantom consisting of water-based scanning gels with inclusions of 3 % w/v agarose gel and emulsion silicone to mimic normal prostate and prostate cancer tissue [120]. The authors found that variability gradually reduced with increasing inclusion size, which could be attributed to the reflection and refraction of the US waves at the boundaries of the objects, resulting in the irregular production of shear waves. The results also demonstrated that, even though there was no significant difference in the SWE value obtained at different depths in most cases, the coefficient of variation demonstrated an increasing tendency with depth, suggesting attenuation of the US beam at a higher depth. The measured SWE values between the machines differed for both phantom materials and were more pronounced in the lower stiffness phantom. Conversely, the intra- and inter-operator agreements

Table 8

Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for the prostate in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Water-based scanning gels Inclusions: Agarose (3 % w/v) gel and emulsion silicone	Prostate	Investigate the factors affecting the variability of transrectal SWE measurements	SWE	N/A	30.08	N/A	Lee <i>et al.</i> , 2023 [120]
PVA (15 wt%) + graphite powder (0.5 wt%) + cellulose particles (0.5 wt%) Inclusion: PVA (20 wt%) + graphite powder (0.5 wt%) Zerdine ® (CIRS, Model 053L)	Tissue-mimicking (stiffness of the inclusions mimic prostate cancer tissue)	Technique testing	Transluminal Elastography, SWE	N/A	3.2	N/A	Gomez <i>et al.</i> , 2021 [125]
Zerdine ® (CIRS, Model 049)	Prostate	Technique validation	S-WAVE	N/A	N/A	N/A	Aleef <i>et al.</i> , 2023 [127]
Zerdine ® (CIRS, Model 049)	Tissue-mimicking	Technique validation	Passive elastography	80	5.1 ± 0.8	N/A	Payen <i>et al.</i> , 2024 [128]
Polyacrylamide: alginate (85:15, 90:10, 95:5 w/w)	Prostate	Mimic pathologic fibrous tissue Evaluate histotripsy treatment protocols	SWE	64.7 ± 25.4	N/A	0.14 ± 0.06	Kumar <i>et al.</i> , 2022 [123]

were excellent.

Beyond the interest in assessing elastography variability and exploring new techniques for the prostate, a recent article [123] focused on developing a phantom that mimicked the tough fibrous tissue characteristic of benign prostatic hyperplasia. Patients with this condition exhibit an enlargement of the prostate gland due to unregulated hyperplastic growth of the epithelial and fibromuscular tissues [124], which can cause increased prostate stiffness. The authors fabricated and characterized the mechanical and acoustic properties of a novel 3D polyacrylamide-alginate phantom hydrogel. Three weight ratios, 85:15, 90:10, and 95:5 w/w (polyacrylamide: alginate), with median Young's moduli measured by SWE of 90.1, 62.7, and 41.3 kPa, respectively, were chosen to represent benign prostatic hyperplasia tissue across its stiffness range. The stiffness was also measured using indentometry techniques, though lower values were obtained, which could result from the differences in the shear rates employed in each technique. They also measured the attenuation and the speed of sound using a through-transmission water substitution method and observed that the acoustic properties had values representative of prostate tissue. Another important characteristic of phantoms for US imaging is acoustic scattering. Under B-mode and SWE, the hydrogel appears to contain acoustic scatterers, thus providing an image with acceptable contrast to US imaging. The results further demonstrated that the hydrogels have a similar response to histotripsy treatment, hence making them a useful, cost-effective alternative for developing and evaluating different treatment protocols.

Most researchers were concerned about fabricating or using commercial phantoms to replicate prostate tissue. Moreover, these phantoms were scanned with transrectal probes to simulate clinical conditions. Notably, there is still potential for developing realistic prostate phantoms that mimic how a patient will be scanned during transrectal elastography by incorporating the prostate gland, urethra, seminal vesicles, rectal wall, and tumors within the phantom. This would guarantee the feasibility of using a technique for prostate elastography.

3.8.2. Material development

Beyond the interest in assessing elastography variability and exploring new techniques for the prostate, a recent article [123] focused on developing a phantom that mimicked the tough fibrous tissue characteristic of benign prostatic hyperplasia. Patients with this condition exhibit an enlargement of the prostate gland due to unregulated hyperplastic growth of the epithelial and fibromuscular tissues [124], which can cause increased prostate stiffness. The authors fabricated and characterized the mechanical and acoustic properties of a novel 3D polyacrylamide-alginate phantom hydrogel. Three weight ratios, 85:15, 90:10, and 95:5 w/w (polyacrylamide: alginate), with median Young's moduli measured by SWE of 90.1, 62.7, and 41.3 kPa, respectively, were chosen to represent benign prostatic hyperplasia tissue across its stiffness range. The stiffness was also measured using indentometry techniques, though lower values were obtained, which could result from the differences in the shear rates employed in each technique. They also measured the attenuation and the speed of sound using a through-transmission water substitution method and observed that the acoustic properties had values representative of prostate tissue. Another important characteristic of phantoms for US imaging is acoustic scattering. Under B-mode and SWE, the hydrogel appears to contain acoustic scatterers, thus providing an image with acceptable contrast to US imaging. The results further demonstrated that the hydrogels have a similar response to histotripsy treatment, hence making them a useful, cost-effective alternative for developing and evaluating different treatment protocols.

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elastography by incorporating the prostate gland, urethra, seminal vesicles, rectal wall, and tumors within the phantom. This would guarantee the feasibility of using a technique for prostate elastography.

3.8.3. Technique development

Gomez et al. [125] presented a new transluminal elastography probe prototype and tested it in phantoms that contained lumen-like conduits and stiffer inclusions. This new technique transmits shear waves directly into the tissue by applying oscillatory rotational forces to the luminal wall. Shear wave interactions with regions of altered tissue elasticity generate reflected waves, a part of which is detected in the lumen [126]. The phantoms of PVA consisted of a softer background (15 wt%) and stiffer inclusions (20 wt%) with stiffness values aligning with observed values in prostate cancer. The stiffness of the inclusions was tuned by increasing the number of freeze-thaw cycles. Graphite powder (0.5 wt %) and cellulose particles (0.5 wt%) were added to increase acoustic scattering. The reflections detected by the transluminal probe were validated against the time-of-flight of the expected reflections calculated from SWV values of the background material and inclusion, measured by SWE. Ultrafast US imaging was used to track the propagation of shear waves emitted by the probe. The prototype detected reflections from the stiffer inclusions within the phantom, and the readouts from the probe showed a good agreement with the time-of-flight estimations.

In another study [127], the authors proposed and characterized a new S-WAVE system for the prostate that can perform 3D volumetric and continuous imaging at the same depth as conventional US without causing tissue or transducer face heating, as opposed to ARF-based methods. A commercial prostate phantom that contains a simulated rectum for transrectal transducers was chosen to characterize the system. This phantom is enclosed in a hard plastic shell with an exposed side that represents the perineum and contains a prostate gland with urethra, seminal vesicles, and three lesions, which are each ≈ 1 cm in diameter. The results showed a statistically significant difference between the measurements of the prostate gland and the lesions, with the lesions having higher stiffness. Furthermore, to validate the measurements, they performed 3D MRE using the same excitation frequencies and the same reconstruction technique, obtaining a high correlation of 96 %.

Payen et al. [128] adapted passive elastography to process B-mode images acquired with a clinical system, to assess elasticity in commercial tissue-mimicking phantoms and detect high-intensity focused ultrasound (HIFU) lesions in prostate cancer patients. Leveraging the physiological wave field *in vivo*, this technique detected significant variations in elasticity before and after HIFU treatment in prostate cancer patients, with an average variation of 33.0 ± 16.0 %.

3.9. Eyes

3.9.1. Motivation

Vision impairment is a major global health issue linked to a lower quality of life and reduced life expectancy among the middle-aged and elderly population [129]. In 2020, an estimated 596 million people had distance vision impairment worldwide, of whom 43 million were blind [130]. The leading causes of blindness are ophthalmic diseases like cataracts, uncorrected refractive error, glaucoma, age-related macular degeneration, and diabetic retinopathy, all of which cause changes in the biomechanical properties of the ocular tissues [131]. Therefore, assessing these properties is crucial for comprehending the eye's physiology, diagnosing and monitoring ocular diseases, and developing effective treatments.

Current ophthalmic diagnostic tools, such as the slit lamp, OCT, and ophthalmoscopy primarily rely on visualizing anatomical structures to detect abnormal morphologic changes [130]. Unlike these traditional methods, elastography offers a more precise diagnosis by quantifying the elastic properties of ocular tissues through measurements of tissue stiffness.

Table 9

Overview of the research and commercially available phantoms with details of their material, type, purpose, and elastography technique used for the eye in the studies analyzed.

Phantom/Material	Phantom Type	Phantom purpose	Elastography technique	Young Modulus (kPa)	Shear wave speed (m/s)	Acoustic attenuation (α – dB/cm/MHz)	Reference
Silicone rubber (1A:1B)	Tissue-mimicking	Test the technique	OCE with Kelvin-Voigt model	N/A	8.5 ± 1.1	N/A	Ramier <i>et al.</i> , 2019 [133]
Gelatin (3, 5 % w/v) + intralipid powder (0.5 % w/v)	Tissue-mimicking	Technique validation	Rev3D-OCE	9.69 ± 8.1	3 ± 1	N/A	Zvietovich <i>et al.</i> , 2019 [134]
Gelatin (7, 12 % w/v) + cellulose (1 % w/v)	Eye	Test the technique	HFUSE	N/A	2.14	N/A	Weng <i>et al.</i> , 2021 [135]
Gelatin + silicon carbide powder + sound scatters	Tissue-mimicking	Technique validation	HFUSE with a mechanical shaker	38.70 ± 5.60	3.38	N/A	Li <i>et al.</i> , 2023 [136]

USE and OCE are high-resolution imaging techniques used in ophthalmology, each with distinct spatial-scale and penetration depth characteristics. USE can generate elasticity maps with millimeter-range resolution at depths of approximately 1–7 cm, while OCE offers better spatial resolution ($<10 \mu\text{m}$) but at a limited penetration depth (of 1–10 mm), reducing its ability to image the entire eye simultaneously [132]. Table 9 summarises the studies that have investigated these techniques in the last 5 years.

Ramier *et al.* [133] opted to use OCE to assess the mechanical properties of the cornea. The paper describes the usage of a point-contact probe driven with various waveforms to generate broadband mechanical waves and analyze their propagation in the cornea. Their system was tested in non-specific silicone rubber phantoms and *ex vivo* porcine eyes to demonstrate reproducibility and test inter-sample variability. Zvietovich *et al.* [134] proposed a reverberant 3D OCE (Rev3D-OCE) to detect elasticity gradients laterally and along the depth of the cornea. The accuracy of this newly proposed system in estimating SWV was validated in tissue-mimicking gelatin phantoms and *ex vivo* porcine cornea. Although both studies successfully characterized the elasticity of the cornea, since neither the type of technique nor the type of phantom is of interest to the present work, the details won't be delved into.

3.9.2. Technique development

Regarding USE, two studies stand out. Weng *et al.* [135] developed an HFUSE method using ultra-fast imaging to assess corneal elasticity. An external vibrator generated elastic waves, with a 40-MHz array transducer capturing the wave propagation. Testing was conducted on 2-mm-thick thin-layer gelatin phantoms and fresh porcine corneas with artificial sclerosis. The phantom was designed to replicate a real cornea, with a liquid-filled anterior chamber, and was immersed in saline externally to simulate typical conditions during US examination. Mechanical tests provided standard elasticity values, which closely matched those from the Lamb wave model used in HFUSE. Due to the high resolution of HFUSE, elastic wave propagation was observed in the porcine cornea. More recently, Li *et al.* [136] developed a high-resolution USE system that uses a mechanical shaker as the vibration source and a moderate frequency range linear array to accurately characterize the biomechanical properties of the whole eye at once. The system was first validated on tissue-mimicking gelatin phantoms and further tested in rabbits with elevated intraocular pressure to understand the influence of intraocular pressure changes on the biomechanical properties of each ocular tissue. The results showed a significant variation in elastic wave speed among different ocular tissues with changing intraocular pressure. The cornea, iris, and optic nerve head showed a linear response, while the lens remained unaffected by changes in intraocular pressure.

A significant deficiency in phantoms that accurately replicate the eye's anatomical, mechanical, and acoustic properties is evident, resulting in a pressing need to rely on *ex vivo* and animal models for validation purposes. Intrinsically, future research in eye elastography

should be focused on developing reliable eye phantoms to test and validate new techniques.

4. Discussion

We provided an overview of papers about the current state of phantoms used for the latest developments in USE technology for vascular, kidney, brain, heart, muscle and tendon, liver, breast, prostate, and eye applications. Through this review, it was possible to verify that gelatin, agar, and PVA-C were the most common materials used across the compiled papers. The use of commercially available phantoms was also quite representative.

When considering acoustic properties such as speed of sound and attenuation, gelatin and agar are the materials that most closely mimic biological tissue and are likely to produce highly realistic US images when used to build anatomical structures. However, both their mechanical properties and longevity are quite limited, demanding strict protocols of handling and storage [6,137]. On the other hand, PVA-C, while also requiring hydration, has better longevity and stability than the previous materials. Another characteristic of interest of this material is that its properties can be tailored through the change of freeze–thaw cycles. In fact, the increase in the number of freeze–thaw cycles is correlated to the increase in the resultant material stiffness [6,10,11,13,137]. This potentiates its use on different organs; for example, it was noted that for the fabrication of vascular phantoms, the use of 6 cycles was common, whilst for prostate tumors, only 2 cycles were used. However, the process is time-consuming, as it requires freezing and slow thawing over 12 h, which can increase the risk of contamination [10,11,13]. Besides this, it is also relevant to mention that Sharma *et al.* noted that the viscosity of the PVA phantom is also significantly affected in this process, which can cause significant errors in USE measurements conducted using linear elastic assumption [13].

It was also possible to observe a lack of information when it comes to the timeline of the realized experiments, as well as the methods used for conserving the phantoms. This lack of information may lead to unreliable results, since, as was previously stated, water-based materials, such as the most commonly used ones, are quite sensitive to environmental factors. In fact, in the case of gelatin, it is known that the refrigeration duration increases the elastic modulus when going from 2 h to 42 h, after which the elastic modulus decreases. However, other factors are also known to alter the properties of the materials, such as the temperature and the way they are stored [6,101,137,138]. Additionally, the fabrication process lacked clarity, which is crucial as slight changes can significantly impact the phantom final properties. As was shown by McIlvain *et al.* [36], phantom homogeneity can be influenced by cooling rate and liquid temperature, while stiffness depends on final liquid temperature and stir rate.

Commercial phantoms are designed for specific applications with limited variation in the anatomical or pathological representation of the targeted structures, often reflecting broad market demands. Their

Young's modulus values are insufficient to capture the full spectrum encountered in clinical settings and, more often than not, fail to provide the adequate lesion-to-background elasticity ratios encountered in real life. Besides, these can cost up to thousands of dollars, and due to their proprietary nature, a full database of constituent materials and their properties can not be accessed. Their ability to mimic various tissue types and build anthropomorphic phantom test devices is still quite limited [7,10,11,137].

Overall, it was evident that, in most cases, the chosen phantoms were rarely the best ones. In fact, there was a clear lack of care to choose materials that adequately mimicked the properties of the anatomical structure under study, opting instead to use "tissue-mimicking" phantoms, which led to the need to supplement the obtained results with *ex/in vivo* experiments. Some exceptions to this were the studies associated with blood vessels, where in all the analyzed papers, the fabricated phantom had the clear intention of representing that structure. In the case of liver and breast studies, a reduced reliance on generic phantoms was observed. This can be attributed to the extensive research on elastography techniques for these organs, which has led to the development of commercially available phantoms that claim to represent their characteristics accurately.

A significant limitation of the chosen phantoms is their isotropic and homogeneous nature, which contrasts with the anisotropic and viscoelastic properties of most human tissues. By neglecting these complexities, the phantom fails to accurately represent the target tissue, potentially compromising the reliability of the obtained results. Furthermore, the human body is composed of multiple layers, including muscles, tendons, ligaments, fascia, fat, fibrous tissue, synovial membranes, nerves, and blood vessels. These layers are dynamic and influence wave propagation through the body, so a simplified, passive, and homogeneous representation of the body may introduce errors in the analysis [8,13,137,139].

Not only do phantoms often fail to mimic anatomical properties, but current elastography techniques also neglect the effects of viscosity and anisotropy, leading to biased elasticity measurements. In essence, to improve tissue characterization and provide a more reliable diagnosis, researchers have been implementing innovative methods to assess viscosity, mainly in the liver, and developing new elastography approaches that consider the anisotropic nature of muscles and tendons. Two other significant technological advancements in this field include the development of 3D elastography and HFUSE, ideal for imaging structures like the eye and heart, which comprise multiple thin layers of tissue. It is noteworthy that despite the established utility of MRE and OCE for studying the brain and eye, USE stands out as a promising modality with potential for further refinement.

In parallel, artificial intelligence has experienced exponential growth in medical image analysis [140], and even though only one of the analyzed studies used artificial intelligence for breast lesions classification [116], its growing use in USE offers significant potential to support clinical decision-making. Additionally, although beyond the scope of this review, digital phantoms emerge as an intermediate assessment tool to validate new elastography approaches and identify potential sources of bias [141]. Moreover, these can be developed without the cost of manufacturing and tailored to specific geometric and material properties [7].

Throughout this review, it was possible to observe the continuous evolution and diversification of elastography approaches, with each research group and manufacturer developing unique technological solutions. It is then clear that measurements vary across different studies, underscoring the need for caution when comparing results. In fact, variable results between systems that use similar techniques but have different manufacturers and even between probes within the same system have been reported [9,142]. Furthermore, the lack of standardized units for measurement hinders the direct comparison of results across different techniques. Another source of variability is viscosity [1,101]. As previously mentioned, the SWV varies with frequency, consequently,

to ensure accurate comparisons of measurements between different modalities, the excitation frequency must be precisely known. Recognizing the importance of understanding the factors influencing elastography measurements, several research groups have conducted studies to investigate the consistency of measurements across different systems. The amount of pressure applied by the US probe is also a critical factor to consider when performing strain and SWE, as demonstrated for the breast, thyroid, and muscle [143–145]. As precompression increases, SE elastogram will be mostly noise while SWV can increase significantly in SWE [146]. While this factor was acknowledged in several studies, with experienced operators conducting the scans, using US probe supports, or scanning phantoms sideways to minimize strain and compression, this information is frequently overlooked.

Accordingly, an essential step toward advancing the field of elastography involves understanding the sources of measurement variability through parametric studies in calibrated phantoms [96]. In that regard, future studies should prioritize the development of standardized phantoms that accurately mimic the anatomy and properties of the different organs and tissues. In US, for instance, some existing standards like the ones proposed by the IEC and the QIBA are already in use and offer some values of interest for the creation of TMMs. However, the studies reviewed did not always mention following these guidelines, and newer modalities demand updated specifications [7,10,96]. For validation of current ultrasound and elastography equipment, standardized tissue-mimicking materials with simple phantoms are sufficient to ensure reproducibility and alignment with IEC-type standards; however, to adapt ultrasound technologies beyond imageology towards therapeutic applications, more anatomically realistic, layered, and anthropomorphic phantoms will likely be required, a twofold standardization pathway that is increasingly being addressed by ongoing QIBA and IEC efforts [147]. Standardization ensures that the materials used in phantoms are rigorously evaluated, enabling the possibility of consistent validation of new approaches and the adequate training of specialists for clinical practice. This consistency is vital for the progress in the design and application of these materials, as well as the advancement of imaging and therapeutic technologies [8].

Furthermore, continued investment in technologies capable of producing materials with microstructures that closely mimic those found in natural soft tissues is imperative, as their microstructures heavily influence the dynamic behavior of soft tissues [8]. Several studies on muscle tissue have already explored the use of various 3D printing techniques to develop anisotropic phantoms, yielding varying degrees of success [57,58]. Given the absence of a definitive optimal manufacturing technique, greater focus and research in this area are warranted. A potentially promising alternative is the employment of 4D printing methods since it offers the potential for dynamic, stimulus-responsive structures that more accurately represent organs [8,148]. The better organ mimicry accomplished, the more reliable the results obtained through elastography are, and the less dependent on *ex vivo* and *in vivo* testing they become.

Indeed, the use of 4D printing methods enables the development of tailored phantoms that more accurately replicate the patient's anatomical structure [148]. As previously stated, acoustic properties vary not only between individuals but also within one's body [6,137]. Real tissues exhibit significant variability in acoustic and mechanical properties due to biological factors such as age, disease, state (*in vivo* or *ex vivo*), and ethnicity [7,137]. Taking this into consideration, there is a clear need to shift the focus from generic phantoms to more specialized designs.

A recurring issue observed in this review is the lack of standardization in phantom fabrication and testing protocols. Variability in experimental timelines, storage conditions, and fabrication processes leads to inconsistent results, limiting the comparability of studies. Furthermore, the isotropic and homogeneous nature of many phantoms contrasts starkly with the anisotropic and viscoelastic properties of real tissues, compromising the reliability of elastography measurements. These

shortcomings in phantom realism have direct clinical implications: insufficiently representative phantoms may lead to over- or under-estimation of tissue stiffness, reducing diagnostic accuracy, and creating barriers for regulatory approval of new elastography systems. Therefore, the development of standardized, anatomically accurate phantoms is not only a scientific need but also a prerequisite for reliable clinical translation and device certification.

Elastography has emerged as a powerful, non-invasive modality capable of quantifying biomechanical tissue properties that are otherwise difficult to assess. In the musculoskeletal domain, more specifically on lower-limb orthoses, most outcomes (e.g., cartilage volume, pain indices) remain subjective or indirect, emphasizing the need for objective biomechanical metrics [149,150]. Together, these works illustrate that elastography, and specifically shear wave elastography, offers a promising avenue for directly measuring changes in cartilage stiffness or structure. By providing reliable, quantitative biomechanical parameters of tissues under orthotic or surgical interventions, elastography could transform our ability to monitor therapeutic efficacy and advance device development.

It is evident from Fig. 3 that there remains a lack of uniform parameterization across the studies reviewed, with several reports omitting fundamental values such as Young's modulus, shear wave velocity, or acoustic attenuation, or reporting them in incompatible ways. This heterogeneity complicates direct comparison of results and limits the reproducibility of elastography validation studies. With the growing adoption of more advanced modelling frameworks and simulation-based validations, it becomes increasingly pertinent that new phantoms are characterized using more rigorous and standardized methodologies. Recent work has shown that advanced approaches, such as explicit time-domain modelling of ultrasound propagation with experimentally validated parameters, can capture frequency-dependent attenuation and nonlinear effects with far higher fidelity than traditional simplifications [151]. Incorporating such techniques into phantom development will not only improve parameter consistency but also strengthen the translational potential of elastography devices.

To ensure meaningful progress, future research should prioritize the development of robust standards for phantom design and USE methodologies. Collaboration between researchers, manufacturers, and regulatory bodies is essential to establish guidelines that address the variability in acoustic and mechanical properties across different organs and individuals. By focusing on the creation of patient-specific, anatomically accurate phantoms, the field can reduce dependence on ex vivo and in vivo testing, paving the way for more reliable and precise diagnostic and therapeutic applications.

Ultimately, continued investment in innovative materials and technologies will be crucial for advancing the field of elastography. The ability to produce phantoms that accurately represent the complex, variable nature of human tissues will not only improve the reliability of elastography techniques but also contribute to the broader goal of enhancing patient care through more accurate imaging and diagnosis.

5. Concluding Remarks

USE has emerged as a promising technique for assessing the mechanical properties of soft tissues. The rapid growth and expanding clinical applications of this technique have prompted the scientific community to develop phantoms capable of providing comprehensive assessments and validation prior to clinical implementation.

This review provided insights into the current state of elastography research as well as the phantoms currently being used for such. A broad spectrum of phantoms was identified, including commercial and research-developed models, though most were unable to accurately mimic the properties of target tissues and organs, leading to a heavy reliance on in vivo and ex vivo studies. It was, however, clear that there's an ever-existing interest in the improvement of elastography techniques. While advancements in this field continue to grow, driven by the

ongoing contributions of multidisciplinary scientists, the development of more realistic phantoms is of utmost importance. Phantom improvements must ensure adequate mechano-acoustic behavior and replication of the physical and functional characteristics of various tissues and organs in both healthy and diseased states.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Shiina T, et al. WFUMB guidelines and Recommendations for Clinical use of Ultrasound Elastography: Part 1: basic Principles and Terminology. *Ultrasound Med Biol* 2015;41(5):1126–47. <https://doi.org/10.1016/j.ultrasmedbio.2015.03.009>.
- [2] J. Ormachea and K. J. Parker, "Elastography imaging: the 30 year perspective," *Phys Med Biol*, vol. 65, no. 24, p. 24TR06, 2020, doi: 10.1088/1361-6560/abca00.
- [3] A. Kumar, K. M. Kempster, M. J. Kerensky, S. Sankar, N. V. Thakor, and A. Manbachi, "Visualizing tactile feedback: an overview of current technologies with a focus on ultrasound elastography," *Front Med Technol*, vol. 5, 2023, doi: 10.3389/fmedt.2023.1238129.
- [4] C. F. Dietrich et al., "EFSUMB Guidelines and Recommendations on the Clinical Use of Liver Ultrasound Elastography, Update 2017 (Long Version)," Aug. 01, 2017, *Georg Thieme Verlag*. doi: 10.1055/s-0043-103952.
- [5] M. Cè et al., "Ultrasound Elastography: Basic Principles and Examples of Clinical Applications with Artificial Intelligence—A Review," Mar. 01, 2023, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/biomedinformatics3010002.
- [6] Armstrong SA, Jafari R, Forsythe JS, Gregory SD. Tissue-mimicking materials for ultrasound-guided needle intervention phantoms: a comprehensive review. *Ultrasound Med Biol* 2023;49(1):18–30. <https://doi.org/10.1016/j.ultrasmedbio.2022.07.016>.
- [7] C. K. McGarry et al., "Tissue mimicking materials for imaging and therapy phantoms: a review," *Phys Med Biol*, vol. 65, no. 23, p. 23TR01, 2020, doi: 10.1088/1361-6560/abbd17.
- [8] Cao Y, Li G-Y, Zhang X, Liu Y-L. Tissue-mimicking materials for elastography phantoms: a review. *Extreme Mech Lett* 2017;17:62–70. <https://doi.org/10.1016/j.eml.2017.09.009>.
- [9] Wang Y, Ono S, Johnson MP, Larson NB, Lynch T, Urban MW. Evaluating variability of commercial liver fibrosis elastography phantoms. *Ultrasound Med Biol* 2023;49(4):1018–30. <https://doi.org/10.1016/j.ultrasmedbio.2022.12.017>.
- [10] Courmane S, Fagan AJ, Browne JE. Review of ultrasound elastography quality control and training test phantoms. *Ultrasound Dec*. 2011;20(1):16–23. <https://doi.org/10.1258/ult.2011.011033>.
- [11] Groves L, et al. "A review of low-cost ultrasound compatible phantoms," *IEEE Trans Biomed Eng*, vol. PP Jun. 2023. <https://doi.org/10.1109/TBME.2023.3288071>.
- [12] A. Jawli, W. Aldehani, G. Nabi, and Z. Huang, "Tissue-Mimicking Material Fabrication and Properties for Multiparametric Ultrasound Phantoms: A Systematic Review," Jun. 01, 2024, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/bioengineering11060620.
- [13] Sharma A, et al. Characterizing viscoelastic polyvinyl alcohol phantoms for ultrasound elastography. *Ultrasound Med Biol* 2023;49(2):497–511. <https://doi.org/10.1016/j.ultrasmedbio.2022.09.019>.

- [14] A. A. Oglat and T. Abukhalil, "Ultrasound Elastography: Methods, Clinical Applications, and Limitations: A Review Article," May 01, 2024, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/app14104308.
- [15] Ozturk A, Grajo JR, Dhyanani M, Samir AE. Principles of ultrasound elastography. *Abdominal Radiology* 2018;43(4):773–85. <https://doi.org/10.1007/s00261-018-1475-6>.
- [16] Pruijssen JT, de Korte CL, Voss I, Hansen HHG. Vascular shear wave elastography in atherosclerotic arteries: a systematic review. *Ultrasound Med Biol* 2020;46(9): 2145–63. <https://doi.org/10.1016/j.ultrasmedbio.2020.05.013>.
- [17] Bontzos G, Douglas V, Douglas K, Kapsala Z, Drakonaki E, Detorakis E. Ultrasound elastography in ocular and periocular tissues: a review. *Current Medical Imaging Formerly Current Medical Imaging Reviews* Jan. 2020;16. <https://doi.org/10.2174/1573405616666201214123117>.
- [18] Liao J, Yang H, Yu J, Liang X, Chen Z. Progress in the application of ultrasound elastography for brain diseases. *J Ultrasound Med Nov.* 2020;39(11):2093–104. <https://doi.org/10.1002/jum.15317>.
- [19] Drakonaki E, Allen G, Wilson D. Ultrasound elastography for musculoskeletal applications. *Br J Radiol* Jan. 2012;85:1435–45. <https://doi.org/10.1259/bjr/93042867>.
- [20] F. Fernandes et al., "Role of Shear Wave Elastography in the Assessment of Myocardial Stiffness in Various Cardiomyopathies," vol. 36, no. 4, p., Dec. 2023, doi: 10.36660/abcmg.202301081.
- [21] X.-W. Cui et al., "Ultrasound elastography," *Endosc Ultrasound*, vol. 11, no. 4, 2022.
- [22] Flaherty ML, et al. Carotid artery stenosis as a cause of stroke. *Neuroepidemiology* Oct. 2012;40(1):36–41. <https://doi.org/10.1159/000341410>.
- [23] Pruijssen JT, Fekkes S, Menssen J, de Korte CL, Hansen HHG. Extending arterial stiffness assessment along the circumference using beam-steered ARFI and wave-tracking: a proof-of-principle study in phantoms and ex vivo. *Comput Struct Biotechnol J* 2023;21:4288–300. <https://doi.org/10.1016/j.csbj.2023.08.024>.
- [24] Pruijssen JT, Schreuder FHBM, Wilbers J, Kaanders JHAM, de Korte CL, Hansen HHG. Performance evaluation of commercial and non-commercial shear wave elastography implementations for vascular applications. *Ultrasonics* 2024; 140:107312. <https://doi.org/10.1016/j.ultras.2024.107312>.
- [25] Chayer B, et al. Atherosclerotic carotid bifurcation phantoms with stenotic soft inclusions for ultrasound flow and vessel wall elastography imaging. *Phys Med Biol* May 2019;64(9):95025. <https://doi.org/10.1088/1361-6560/ab1145>.
- [26] King DM, Moran CM, McNamara JD, Fagan AJ, Browne JE. Development of a vessel-mimicking material for use in anatomically realistic doppler flow phantoms. *Ultrasound Med Biol* May 2011;37(5):813–26. <https://doi.org/10.1016/j.ultrasmedbio.2011.02.012>.
- [27] Malone AJ, Cournane S, Naydenova IG, Fagan AJ, Browne JE. Polyvinyl alcohol cryogel based vessel mimicking material for modelling the progression of atherosclerosis. *Phys Med Jan.* 2020;69:1–8. <https://doi.org/10.1016/j.ejomp.2019.11.012>.
- [28] Ramnarine KV, Garrard JW, Kanber B, Nduwayo S, Hartshorne TC, Robinson TG. Shear wave elastography imaging of carotid plaques: feasible, reproducible and of clinical potential. *Cardiovasc Ultrasound* Dec. 2014;12(1):49. <https://doi.org/10.1186/1476-7120-12-49>.
- [29] Ramnarine KV, Garrard JW, Dexter K, Nduwayo S, Panerai RB, Robinson TG. Shear wave elastography assessment of carotid plaque stiffness: in vitro reproducibility study. *Ultrasound Med Biol Jan.* 2014;40(1):200–9. <https://doi.org/10.1016/j.ultrasmedbio.2013.09.014>.
- [30] Li H, Porée J, Roy Cardinal M-H, Cloutier G. Two-dimensional affine model-based estimators for principal strain vascular ultrasound elastography with compound plane wave and transverse oscillation beamforming. *Ultrasonics* 2019;91:77–91. <https://doi.org/10.1016/j.ultras.2018.07.012>.
- [31] Li H, Porée J, Chayer B, Cardinal M-H-R, Cloutier G. Parameterized strain estimation for vascular ultrasound elastography with sparse representation. *IEEE Trans Med Imaging* 2020;39(12):3788–800. <https://doi.org/10.1109/TMI.2020.3005017>.
- [32] Li Y, Liu Y, Gao L, Tian C. Renal stiffness measured by shear wave elastography and its relationship with perirenal fat in patients with chronic kidney disease. *J Clin Ultrasound* Jan. 2024;52(1):3–12. <https://doi.org/10.1002/jcu.23598>.
- [33] Kijanka P, Vasconcelos L, Mandrekar J, Urban MW. Evaluation of robustness of S-transform based phase velocity estimation in viscoelastic phantoms and renal transplants. *IEEE Trans Biomed Eng* 2024;71(3):954–66. <https://doi.org/10.1109/TBME.2023.3323983>.
- [34] Leong SS, et al. Stiffness and anisotropy effect on shear wave elastography: a phantom and in vivo renal study. *Ultrasound Med Biol* 2020;46(1):34–45. <https://doi.org/10.1016/j.ultrasmedbio.2019.08.011>.
- [35] Kishimoto R, et al. Intra- and inter-operator reproducibility of US point shear-wave elastography in various organs: evaluation in phantoms and healthy volunteers. *Eur Radiol* 2019;29(11):5999–6008. <https://doi.org/10.1007/s00330-019-06195-8>.
- [36] McIlvain G, Ganji E, Cooper C, Killian ML, Ogunnaike BA, Johnson CL. Reliable preparation of agarose phantoms for use in quantitative magnetic resonance elastography. *J Mech Behav Biomed Mater* 2019;97:65–73. <https://doi.org/10.1016/j.jmbbm.2019.05.001>.
- [37] Zhu Y, et al. Supershear rayleigh wave imaging for quantitative assessment of biomechanical properties of brain using air-coupled optical coherence elastography. *APL Bioeng* Jan. 2023;7. <https://doi.org/10.1063/5.0160213>.
- [38] Lay F-Y, Chen P-Y, Cheng H-F, Kuo Y-M, Huang C-C. Ex vivo evaluation of mouse brain elasticity using high-frequency ultrasound elastography. *IEEE Trans Biomed Eng* 2019;66(12):3426–35. <https://doi.org/10.1109/TBME.2019.2905551>.
- [39] Haupt RW, Fenn AJ, Brigada DJ, Laher RM, Samir AE. "Noninvasive transcranial ultrasound system," in. *IEEE International Ultrasonics Symposium (IUS)* 2022; 2022:1–6. <https://doi.org/10.1109/IUS54386.2022.9957460>.
- [40] Zhu Y, et al. Noncontact longitudinal shear wave imaging for the evaluation of heterogeneous porcine brain biomechanical properties using optical coherence elastography. *Biomed Opt Express* Jan. 2023;14. <https://doi.org/10.1364/BOE.497801>.
- [41] N. Li, P. Gaur, K. Quah, and K. Butts Pauly, "Improving in situ acoustic intensity estimates using MR acoustic radiation force imaging in combination with multifrequency MR elastography," *Magn Reson Med*, vol. 88, no. 4, pp. 1673–1689, Oct. 2022, doi: <https://doi.org/10.1002/mrm.29309>.
- [42] Tardieu M, et al. Magnetic resonance elastography with guided pressure waves. *NMR Biomed* Jul. 2022;35(7):e4701.
- [43] Du B, et al. A novel transcranial ultrasound imaging method with diverging wave transmission and deep learning approach. *Comput Methods Programs Biomed* 2020;186:105308. <https://doi.org/10.1016/j.cmpb.2019.105308>.
- [44] Yin Z, Li G-Y, Zhang Z, Zheng Y, Cao Y. SWENet: a physics-informed deep neural network (PINN) for shear wave elastography. *IEEE Trans Med Imaging* 2024;43(4):1434–48. <https://doi.org/10.1109/TMI.2023.3338178>.
- [45] M. Yushchenko, M. Sarraçanie, and N. Salameh, "Fast acquisition of propagating waves in humans with low-field MRI: Toward accessible MR elastography," *Sci Adv*, vol. 8, no. 36, p. eabo5739, 2022, doi: 10.1126/sciadv.abo5739.
- [46] Du B, Zheng H, Fang S, Chen S, Lu M, Mao R. A novel transcranial ultrasound imaging method with diverging wave. In: *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*; 2019. p. 6640–3. <https://doi.org/10.1109/EMBC.2019.8856896>.
- [47] Pedreira O, et al. "Smart ultrasound device for non-invasive real-time myocardial stiffness quantification of the human heart," *IEEE Trans Biomed Eng*, vol. PP Jan. 2021. <https://doi.org/10.1109/TBME.2021.3087039>.
- [48] Urban MW, Qiang B, Song P, Nenadic IZ, Chen S, Greenleaf JF. Investigation of the effects of myocardial anisotropy for shear wave elastography using impulsive force and harmonic vibration. *Phys Med Biol* Jan. 2016;61(1):365–82. <https://doi.org/10.1088/0031-9155/61/1/365>.
- [49] Mishra A, Cleveland R. Agarose as a tissue mimic for the porcine heart, kidney, and liver: measurements and a springpot model. *Bioengineering* Jan. 2024;11: 589. <https://doi.org/10.3390/bioengineering11060589>.
- [50] Lee S, Eun L, Hwang J, Eun Y. Ex vivo evaluation of mechanical anisotropic tissues with high-frequency ultrasound shear wave elastography. *Sensors* Jan. 2022;22:978. <https://doi.org/10.3390/s22030978>.
- [51] Chen P-Y, Yang T-H, Kuo L-C, Shih C-C, Huang C-C. Characterization of hand tendons through high-frequency ultrasound elastography. *IEEE Trans Ultrason Ferroelectr Freq Control* 2020;67(1):37–48. <https://doi.org/10.1109/TUFFC.2019.2938147>.
- [52] Chen P-Y, Yang T-H, Kuo L-C, Hsu H-Y, Su F-C, Huang C-C. Evaluation of hand tendon elastic properties during rehabilitation through high-frequency ultrasound shear elastography. *IEEE Trans Ultrason Ferroelectr Freq Control* Aug. 2021;68(8):2716–26. <https://doi.org/10.1109/TUFFC.2021.3077891>.
- [53] Xu G-X, Chen P-Y, Huang C-C. Visualization of human hand tendon mechanical properties in 3-D using high-frequency dual-direction shear wave imaging. *IEEE Trans Ultrason Ferroelectr Freq Control* 2023;70(11):1457–69. <https://doi.org/10.1109/TUFFC.2023.3312273>.
- [54] Götschi T, Schulz N, Snedeker J, Hanemann J, Franchi M, Spörri J. Three-dimensional mapping of shear wave velocity in human tendon: a proof of concept study. *Sensors* Jan. 2021. <https://doi.org/10.3390/s21051655>.
- [55] Ngo H-H-P, et al. In plane quantification of in vivo muscle elastic anisotropy factor by steered ultrasound pushing beams. *Phys Med Biol* 2024;69(4):045013. <https://doi.org/10.1088/1361-6560/ad21a0>.
- [56] Dong J, Zhao J, Liu X, Lee W-N. Nondestructive ultrasound evaluation of microstructure-related material parameters of skeletal muscle: an in silico and in vitro study. *J Mech Behav Biomed Mater* Jan. 2023;142:105807. <https://doi.org/10.1016/j.jmbbm.2023.105807>.
- [57] Guidetti M, et al. Anisotropic composite material phantom to improve skeletal muscle characterization using magnetic resonance elastography. *J Mech Behav Biomed Mater* 2019;89:199–208. <https://doi.org/10.1016/j.jmbbm.2018.09.032>.
- [58] H.-J. Chen, H.-C. Chuang, G.-X. Xu, C. Chen, W.-R. Su, and C.-C. Huang, "Wearable Ultrasound Imaging Device for Dynamic Dual-Direction Shear Wave Elastography of Shoulder Muscle," *IEEE Trans Ultrason Ferroelectr Freq Control*, vol. PP, p. 1, Jan. 2024, doi: 10.1109/TUFFC.2023.3348472.
- [59] Bied M, Jourdain L, Genisson J-L. "Acoustoelasticity in transverse isotropic soft tissues: quantification of muscles' nonlinear elasticity," in. *IEEE International Ultrasonics Symposium (IUS)* 2020;2020:1–4. <https://doi.org/10.1109/IUS46767.2020.9251574>.
- [60] Fernandes NATC, et al. Sustainable tunable anisotropic ultrasound medical phantoms for skin, skeletal muscle, and other fibrous biological tissues using natural fibers and a bio-elastomeric matrix. *Journal of Composites Science* Jul. 2025;9(7):370. <https://doi.org/10.3390/jcs9070370>.
- [61] Tsuchida W, et al. Application of the novel estimation method by shear wave elastography using vibrator to human skeletal muscle. *Sci Rep* 2020;10(1):22248. <https://doi.org/10.1038/s41598-020-79215-z>.
- [62] Marcellin P, Kutala BK. Liver diseases: a major, neglected global public health problem requiring urgent actions and large-scale screening. *Liver Int* Feb. 2018; 38(S1):2–6. <https://doi.org/10.1111/liv.13682>.
- [63] Younossi ZM, Wong G, Anstee QM, Henry L. The Global Burden of Liver Disease. *Clin Gastroenterol Hepatol* 2023;21(8):1978–91. <https://doi.org/10.1016/j.cgh.2023.04.015>.

- [64] Dhyani M, Anvari A, Samir AE. Ultrasound elastography: liver. *Abdom Imaging* 2015;40(4):698–708. <https://doi.org/10.1007/s00261-015-0373-4>.
- [65] G. Suffredini, W. D. Gao, and J. M. Dodd-o, “Ultrasound Shear Wave Elastography Evaluation of the Liver and Implications for Perioperative Medicine,” Jul. 01, 2024, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/jcm13133633.
- [66] N. Korta Martiartu et al., “Sources of Variability in Shear Wave Speed and Dispersion Quantification with Ultrasound Elastography: A Phantom Study,” *Ultrasound Med Biol*, vol. 47, no. 12, pp. 3529–3542, Dec. 2021, doi: 10.1016/j.ultrasmedbio.2021.08.013.
- [67] Hwang JA, Jeong WK, Song KD, Kang KA, Lim HK. 2-D shear wave elastography for focal lesions in liver phantoms: effects of background stiffness, depth and size of focal lesions on stiffness measurement. *Ultrasound Med Biol Dec*. 2019;45(12): 3261–8. <https://doi.org/10.1016/j.ultrasmedbio.2019.08.006>.
- [68] Moalla S, et al. Methodological study to investigate the potential of ultrasound-based elastography and texture as biomarkers to monitor liver tumors. *Diagnostics* 2020;10(10):Oct. <https://doi.org/10.3390/diagnostics10100811>.
- [69] Racedo J, Urban MW. Evaluation of reconstruction parameters for 2-D comb-push ultrasound shear wave elastography. *IEEE Trans Ultrason Ferroelectr Freq Control* Feb. 2019;66(2):254–63. <https://doi.org/10.1109/TUFFC.2018.2884348>.
- [70] Z. H. Huang et al., “Palm-Sized Wireless Transient Elastography System with Real-Time B-Mode Ultrasound Imaging Guidance: Toward Point-of-Care Liver Fibrosis Assessment,” *Diagnostics*, vol. 14, no. 2, Jan. 2024, doi: 10.3390/diagnostics14020189.
- [71] Cormack JM, Chao YH, Archer BT, Kim K, Spratt KS, Hamilton MF. Focused shear wave beam propagation in tissue-mimicking phantoms. *IEEE Trans Biomed Eng* Feb. 2024;71(2):621–30. <https://doi.org/10.1109/TBME.2023.3311688>.
- [72] Selladurai S, Thittai AK. Towards quantitative quasi-static ultrasound elastography using a reference layer for liver imaging application: a preliminary assessment. *Ultrasonics* Mar. 2019;93:7–17. <https://doi.org/10.1016/j.ultras.2018.10.007>.
- [73] Selladurai S, Verma A, Thittai AK. Toward quantitative and operator-independent quasi-static ultrasound elastography: an ex vivo feasibility study. *Ultrasound Imaging Jul*. 2020;42(4–5):179–90. <https://doi.org/10.1177/0161734620921532>.
- [74] M. Aboutaleb, N. Kheirkhah, A. Samani, and A. Sadeghi-Naini, “An Enhanced Method for Full-Inversion-Based Ultrasound Elastography of the Liver,” in *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 3887–3890. doi: 10.1109/EMBC48229.2022.9871656.
- [75] Zeng Q, et al. “Multifrequency liver shear wave absolute vibro-elastography with an xmatrix array - 2D vs. 3D Comparison Study,” *2022 IEEE International Ultrasonics Symposium (IUS)*; 2022. p. 1–4.
- [76] Huang C, et al. Three-dimensional shear wave elastography on conventional ultrasound scanners with external vibration. *Phys Med Biol* 2020;65(21):Nov. <https://doi.org/10.1088/1361-6560/aba5ea>.
- [77] Zeng Q, et al. Three-Dimensional Multi-Frequency Shear Wave absolute Vibro-Elastography (3D S-WAVE) with a Matrix Array Transducer: Implementation and Preliminary in Vivo Study of the Liver. *IEEE Trans Med Imaging* Feb. 2021;40(2): 648–60. <https://doi.org/10.1109/TMI.2020.3034065>.
- [78] Hashemi HS, Mohammed SK, Zeng Q, Azar RZ, Rohling RN, Salcudean SE. 3-D ultrafast shear wave absolute vibro-elastography using a matrix array transducer. *IEEE Trans Ultrason Ferroelectr Freq Control* Sep. 2023;70(9):1039–53. <https://doi.org/10.1109/TUFFC.2023.3280450>.
- [79] I. H. Faris, J. Melchor, A. Callejas, J. Torres, and G. Rus, “Viscoelastic biomarkers of ex vivo liver samples via torsional wave elastography,” *Diagnostics*, vol. 10, no. 2, 2020, doi: 10.3390/diagnostics10020111.
- [80] Hofstetter LW, et al. Efficient shear wave elastography using transient acoustic radiation force excitations and MR displacement encoding. *Magn Reson Med* May 2019;81(5):3153–67. <https://doi.org/10.1002/mrm.27647>.
- [81] Lorée H, Bastard C, Miette V, Sandrin L. Vibration-guided transient elastography: a novel fibroscan® examination with improved guidance for liver stiffness measurement. *Ultrasound Med Biol Sep*. 2020;46(9):2193–206. <https://doi.org/10.1016/j.ultrasmedbio.2020.04.010>.
- [82] Czernuszewicz TJ, et al. Development of a robotic shear wave elastography system for noninvasive staging of liver disease in murine models. *Hepatol Commun* 2022;6(7):1827–39. <https://doi.org/10.1002/hep4.1912/supplinfo>.
- [83] Ahmed R, Ye J, Gerber SA, Linehan DC, Doyle MM. Preclinical imaging using single track location shear wave elastography: monitoring the progression of murine pancreatic tumor liver metastasis in vivo. *IEEE Trans Med Imaging Jul*. 2020;39(7):2426–39. <https://doi.org/10.1109/TMI.2020.2971422>.
- [84] P. Kijanka, B. Qiang, P. Song, C. Amador Carrascal, S. Chen, and M. Urban, “Robust Phase Velocity Dispersion Estimation of Viscoelastic Materials Used for Medical Applications Based on the Multiple Signal Classification Method,” *IEEE Trans Ultrason Ferroelectr Freq Control*, vol. 65, pp. 423–439, Jan. 2018, doi: 10.1109/TUFFC.2018.2792324.
- [85] Kijanka P, Urban MW. Two-point frequency shift method for shear wave attenuation measurement. *IEEE Trans Ultrason Ferroelectr Freq Control* 2020;67(3):483–96. <https://doi.org/10.1109/TUFFC.2019.2945620>.
- [86] Kijanka P, Ambrozinski L, Urban MW. Two point method for robust shear wave phase velocity dispersion estimation of viscoelastic materials. *Ultrasound Med Biol Sep*. 2019;45(9):2540–53. <https://doi.org/10.1016/j.ultrasmedbio.2019.04.016>.
- [87] Kijanka P, Urban MW. Phase velocity estimation with expanded bandwidth in viscoelastic phantoms and tissues. *IEEE Trans Med Imaging* May 2021;40(5): 1352–62. <https://doi.org/10.1109/TMI.2021.3054950>.
- [88] Kijanka P, Urban MW. Local phase velocity based imaging of viscoelastic phantoms and tissues. *IEEE Trans Ultrason Ferroelectr Freq Control* Mar. 2021;68(3):389–405. <https://doi.org/10.1109/TUFFC.2020.2968147>.
- [89] Kijanka P, Urban MW. Local phase velocity based imaging: a new technique used for ultrasound shear wave elastography. *IEEE Trans Med Imaging* Apr. 2019;38(4):894–908. <https://doi.org/10.1109/TMI.2018.2874545>.
- [90] R. Koda et al., “Application of continuous shear wave elastography method with multiple frequency selection to liver viscoelasticity measurement,” *Jpn J Appl Phys*, vol. 63, no. 4, Apr. 2024, doi: 10.35848/1347-4065/ad3ae4.
- [91] Bhatt M, et al. Reconstruction of viscosity maps in ultrasound shear wave elastography. *IEEE Trans Ultrason Ferroelectr Freq Control* Jun. 2019;66(6): 1068–78. <https://doi.org/10.1109/TUFFC.2019.2908550>.
- [92] Kijanka P, Urban MW. Improved two-point frequency shift power method for measurement of shear wave attenuation. *Ultrasonics* Aug. 2022;124. <https://doi.org/10.1016/j.ultras.2022.106735>.
- [93] Yazdani L, Bhatt M, Rafati I, Tang A, Cloutier G. The revisited frequency-shift method for shear wave attenuation computation and imaging. *IEEE Trans Ultrason Ferroelectr Freq Control* Jun. 2022;69(6):2061–74. <https://doi.org/10.1109/TUFFC.2022.3166448>.
- [94] Misliti R, Iliza KT, Gerber SA, Doyle MM. Merging single-track location Elastographic imaging with the frequency shift method improves shear wave attenuation measurements. *Front Phys* 2024;12. <https://doi.org/10.3389/fphy.2024.1326770>.
- [95] Khan S, Goswami S, Feng F, Hollenbach S, Doyle MM, McAleavey SA. Probing tissue viscoelasticity with STL ultrasound shearwave spectroscopy using cole-cole diagrams. *IEEE Trans Biomed Eng* Mar. 2024;71(3):916–28. <https://doi.org/10.1109/TBME.2023.3322420>.
- [96] Palmeri ML, et al. Radiological society of north america/quantitative imaging biomarker alliance shear wave speed bias quantification in elastic and viscoelastic phantoms. *J Ultrasound Med Mar*. 2021;40(3):569–81. <https://doi.org/10.1002/jum.15609>.
- [97] Kishimoto R, et al. Shear wave speed measurement bias in a viscoelastic phantom across six ultrasound elastography systems: a comparative study with transient elastography and magnetic resonance elastography. *J Med Ultrason* Apr. 2022;49(2):143–52. <https://doi.org/10.1007/s10396-022-01190-x>.
- [98] Gilligan LA, Trout AT, Bennett P, Dillman JR. Repeatability and agreement of shear wave speed measurements in phantoms and human livers across 6 ultrasound 2-dimensional shear wave elastography systems. *Invest Radiol* Apr. 2020;55(4):191–9. <https://doi.org/10.1097/RLI.0000000000000627>.
- [99] Hwang J, Yoon HM, Jung AY, Lee JS, Cho YA. Comparison of 2-dimensional shear wave elastographic measurements using elastog imaging and supersonic shear imaging: phantom study and clinical pilot study. *J Ultrasound Med* Feb. 2020;39(2):311–21. <https://doi.org/10.1002/jum.15108>.
- [100] Serra JT, Mueller J, Teng H, Elshaarawy O, Mueller S. Prospective comparison of transient elastography using two different devices: performance of fibroscan and fibrotouch. *Hepat Med Mar*. 2020;12:41–8. <https://doi.org/10.2147/hmer.s245455>.
- [101] A. S. Morr et al., “Liquid-liver phantom: mimicking the viscoelastic dispersion of human liver for elastography in ultrasound and MRI.”
- [102] Courane S, Browne JE, Fagan AJ. The effects of fatty deposits on the accuracy of the Fibroscan® liver transient elastography ultrasound system. *Phys Med Biol* Jun. 2012;57(12):3901–14. <https://doi.org/10.1088/0031-9155/57/12/3901>.
- [103] Courane S, Cannon L, Browne JE, Fagan AJ. Assessment of the accuracy of an ultrasound elastography liver scanning system using a PVA-cryogel phantom with optimal acoustic and mechanical properties. *Phys Med Biol* Oct. 2010;55(19): 5965–83. <https://doi.org/10.1088/0031-9155/55/19/022>.
- [104] Sung H, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* May 2021;71(3):209–49. <https://doi.org/10.3322/caac.21660>.
- [105] Arnold M, et al. Current and future burden of breast cancer: Global statistics for 2020 and 2040. *Breast* 2022;66:15–23. <https://doi.org/10.1016/j.breast.2022.08.010>.
- [106] L. Wang, “Early diagnosis of breast cancer,” *Sensors (Switzerland)*, vol. 17, no. 7, Jul. 2017, doi: 10.3390/s17071572.
- [107] Mann RM, Hooley R, Barr RG, Moy L. Novel Approaches to Screening for Breast Cancer. *Radiology* Sep. 2020;297(2):266–85. <https://doi.org/10.1148/radiol.2020200172>.
- [108] Leonov D, et al. Development of an anatomical breast phantom from polyvinyl chloride plastisol with lesions of various shape, elasticity and echogenicity for teaching ultrasound examination. *Int J Comput Assist Radiol Surg* Jan. 2024;19(1):151–61. <https://doi.org/10.1007/s11548-023-02911-4>.
- [109] G. Aldosary et al., “Radiological, dosimetric and mechanical properties of a deformable breast phantom for radiation therapy and surgical applications,” *Biomed Phys Eng Express*, vol. 6, no. 3, May 2020, doi: 10.1088/2057-1976/ab834a.
- [110] Wang Y, Jacobson DS, Urban MW. A Non-invasive method to estimate the stress-strain curve of soft tissue using ultrasound elastography. *Ultrasound Med Biol* May 2022;48(5):786–807. <https://doi.org/10.1016/j.ultrasmedbio.2021.12.016>.
- [111] Rabin C, Benesh N. Quantitative breast elastography from B-mode images. *Med Phys* Jul. 2019;46(7):3001–12. <https://doi.org/10.1002/mp.13537>.
- [112] Matrone G, Lashkevich E, Berton A, Ramalli A, Savoia AS, Magenes G. Effects of coherence-based beamforming on breast ultrasound elastograms. In: in *2019 IEEE*

- International Ultrasonics Symposium (IUS), Oct; 2019. p. 312–5. <https://doi.org/10.1109/ULTSYM.2019.8926022>.
- [113] Caius M, Samani A. Comparative study of ultrasound tissue motion tracking techniques for effective breast ultrasound elastography. *Applied Sciences (Switzerland)* 2023;13(21):Nov. <https://doi.org/10.3390/app13211912>.
- [114] Kelly C, Lobo J, Honarvar M, Shao Y, Salcudean S. “An Automated breast ultrasound scanner with integrated shear wave elastography, doppler flow imaging and photoacoustic tomography; an automated breast ultrasound scanner with integrated shear wave elastography. *Doppler Flow Imaging and Photoacoustic Tomography*” 2018.
- [115] S. Di Meo, L. Pasotti, E. Lashkevich, G. Magenes, M. Pasian, and G. Matrone, “Combining Millimeter-Wave Imaging, Ultrasound and Elastography in a New Multimodal Approach for Breast Cancer Detection: Initial Experimental Results*,” Jan. 2020, pp. 1807–1810. doi: 10.1109/EMBC44109.2020.9176088.
- [116] An J, Abdus-Shakur T, Denis M. Quantitative assessment of tissue stiffness using transfer learning ultrasound elastography: a breast cancer phantom study. *IEEE Sens Lett* 2023;7(10):Oct. <https://doi.org/10.1109/LENS.2023.3307102>.
- [117] Kumar V, et al. Viscoelastic parameters as discriminators of breast masses: initial human study results. *PLoS One* 2018;13(10):Oct. <https://doi.org/10.1371/journal.pone.0205717>.
- [118] Gubarkova EV, et al. Compression optical coherence elastography versus strain ultrasound elastography for breast cancer detection and differentiation: pilot study. *Biomed Opt Express* May 2022;13(5):2859. <https://doi.org/10.1364/boe.451059>.
- [119] Rawla P. Epidemiology of prostate cancer. *World J Oncol* 2019;10(2):63–89. <https://doi.org/10.14740/wjon1191>.
- [120] Lee J, Yoon SK, Cho JH, Kwon HJ, Kim DW, Lee JW. Variability of transrectal shear wave elastography in a phantom model. *Journal of the Korean Society of Radiology* Sep. 2023;84(5):1110–22. <https://doi.org/10.3348/jksr.2023.0051>.
- [121] Almkali YE, et al. Advanced strain elastography is a reliable approach for prostate cancer detection in patients with elevated PSA levels. *Sci Rep* 2024;14(1):2917. <https://doi.org/10.1038/s41598-024-53440-2>.
- [122] C. Luthold, T. Hallal, D. P. Labbé, and F. Bordeleau, “The Extracellular Matrix Stiffening: A Trigger of Prostate Cancer Progression and Castration Resistance?,” Jun. 01, 2022, *MDPI*. doi: 10.3390/cancers14122887.
- [123] Y. Nanda Kumar et al., “Development of Tough Hydrogel Phantoms to Mimic Fibrous Tissue for Focused Ultrasound Therapies,” *Ultrasound Med Biol*, vol. 48, no. 9, pp. 1762–1777, Sep. 2022, doi: 10.1016/j.ultrasmedbio.2022.05.002.
- [124] Devlin CM, Simms MS, Maitland NJ. Benign prostatic hyperplasia – what do we know? *BJU Int* Apr. 2021;127(4):389–99. <https://doi.org/10.1111/bju.15229>.
- [125] A. Gomez, M. Hurtado, A. Callejas, J. Torres, N. Saffari, and G. Rus, “Experimental evidence of generation and reception by a transluminal axisymmetric shear wave elastography prototype,” *Diagnostics*, vol. 11, no. 4, Apr. 2021, doi: 10.3390/diagnostics11040645.
- [126] Gomez A, Rus G, Saffari N. Reverse time migration and genetic algorithms combined for reconstruction in transluminal shear wave elastography: an in silico case study. *Ultrasonics* 2024;138:107206. <https://doi.org/10.1016/j.ultras.2023.107206>.
- [127] Aleef TA, et al. Multi-frequency 3D shear wave absolute vibro-elastography (S-WAVE) system for the prostate. *IEEE Trans Med Imaging* Nov. 2023;42(11):3436–50. <https://doi.org/10.1109/TMI.2023.3288468>.
- [128] Payen T, et al. Passive elastography for clinical HIFU lesion detection. *IEEE Trans Med Imaging* Apr. 2024;43(4):1594–604. <https://doi.org/10.1109/TMI.2023.3344182>.
- [129] M. V. Cicinelli, S. Marmamula, and R. C. Khanna, “Comprehensive eye care - Issues, challenges, and way forward,” *Indian J Ophthalmol*, vol. 68, no. 2, 2020.
- [130] Burton MJ, et al. The lancet global health commission on global eye health: vision beyond 2020. *Lancet Glob Health* Apr. 2021;9(4):e489–551. [https://doi.org/10.1016/S2214-109X\(20\)30488-5](https://doi.org/10.1016/S2214-109X(20)30488-5).
- [131] Zhang J, et al. Elasticity measurements of ocular anterior and posterior segments using optical coherence elastography. *Opt Express* Apr. 2022;30(9):14311–8. <https://doi.org/10.1364/OE.456065>.
- [132] Ormachea J, Zvietcovich F. Reverberant Shear Wave Elastography: a Multi-Modal and Multi-Scale Approach to measure the Viscoelasticity Properties of Soft Tissues. *Front Phys* Jan. 2021;8. <https://doi.org/10.3389/fphy.2020.606793>.
- [133] Ramier A, Tavakol B, Yun S-H. Measuring mechanical wave speed, dispersion, and viscoelastic modulus of the cornea using optical coherence elastography. *Opt Express* Jun. 2019;27(12):16635–49. <https://doi.org/10.1364/OE.27.016635>.
- [134] Zvietcovich F, Pongchalee P, Meemon P, Rolland JP, Parker KJ. Reverberant 3D optical coherence elastography maps the elasticity of individual corneal layers. *Nat Commun* 2019;10(1):Dec. <https://doi.org/10.1038/s41467-019-12803-4>.
- [135] Weng C-C, Chen P-Y, Chou D, Shih C-C, Huang C-C. High Frequency Ultrasound Elastography for estimating the Viscoelastic Properties of the Cornea using Lamb Wave Model. *IEEE Trans Biomed Eng Sep.* 2021;68(9):2637–44. <https://doi.org/10.1109/TBME.2020.3044066>.
- [136] Li R, et al. Simultaneous Assessment of the whole Eye Biomechanics using Ultrasonic Elastography. *IEEE Trans Biomed Eng Apr.* 2023;70(4):1310–7. <https://doi.org/10.1109/TBME.2022.3215498>.
- [137] Culjat MO, Goldenberg D, Tewari P, Singh RS. A Review of Tissue Substitutes for Ultrasound Imaging. *Ultrasound Med Biol* 2010;36(6):861–73. <https://doi.org/10.1016/j.ultrasmedbio.2010.02.012>.
- [138] J. Dahmani, C. Laporte, D. Pereira, P. Belanger, and Y. Petit, “Predictive Model for Designing Soft-Tissue Mimicking Ultrasound Phantoms With Adjustable Elasticity,” *IEEE Trans Ultrason Ferroelectr Freq Control*, vol. PP, p. 1, Jan. 2019, doi: 10.1109/TUFFC.2019.2953190.
- [139] Patey SJ, Corcoran JP. Physics of ultrasound. *Anaesthesia & Intensive Care Medicine* 2021;22(1):58–63. <https://doi.org/10.1016/j.mpaic.2020.11.012>.
- [140] Fang C, Sidhu PS. Ultrasound-based liver elastography: current results and future perspectives. *Abdominal Radiology* 2020;45(11):3463–72. <https://doi.org/10.1007/s00261-020-02717-x>.
- [141] Pasyar P, Masjoodi S, Montazeriani Z, Makkiabadi B. A digital viscoelastic liver phantom for investigation of elastographic measurements. *Comput Biol Med* 2020;127:104078. <https://doi.org/10.1016/j.combiomed.2020.104078>.
- [142] Pelea M-A, Serban O, Bădărină M, Fodor D. Advancements and challenges in Shear-Wave Elastography of tendons: a comprehensive review. *Med Ultrason Jan.* 2024. <https://doi.org/10.11152/mu-4444>.
- [143] Barr RG, Zhang Z. Effects of Precompression on Elasticity Imaging of the Breast. *J Ultrasound Med Jun.* 2012;31(6):895–902. <https://doi.org/10.7863/jum.2012.31.6.895>.
- [144] Lam ACL, Pang SWA, Ahuja AT, Bhatia KSS. The influence of precompression on elasticity of thyroid nodules estimated by ultrasound shear wave elastography. *Eur Radiol* 2016;26(8):2845–52. <https://doi.org/10.1007/s00330-015-4108-2>.
- [145] Wang X, et al. Effect of acquisition depth and precompression from probe and couplant on shear wave elastography in soft tissue: an in vitro and in vivo study. *Quant Imaging Med Surg Jan.* 2020;10:754–65. <https://doi.org/10.21037/qims.2020.01.15>.
- [146] R. G. Barr, *Elastography: A Practical Approach*. Stuttgart: Georg Thieme Verlag KG, 2017. doi: 10.1055/b-0036-140634.
- [147] P. Carson, “MO-D-218-01: Overview of Methodology and Standards (QIBA, IEC, AIUM and AAPM),” *Med Phys*, vol. 39, no. 6Part21, pp. 3869–3870, Jun. 2012, doi: 10.1118/1.4735799.
- [148] Ramezani M, Ripin Z. 4D Printing in Biomedical Engineering: Advancements, challenges, and Future Directions. *J Funct Biomater Jan.* 2023;14:347. <https://doi.org/10.3390/jfb14070347>.
- [149] Fernandes N, Silva F, Carvalho Ó, Leal A. Effect of lower limb orthoses on cartilage in patients with knee osteoarthritis: a narrative review. *Prosthet Orthot Int* Oct. 2022;46(5):466–76. <https://doi.org/10.1097/PXR.000000000000128>.
- [150] Fernandes NATC, Arieira A, Hinckel B, Silva FS, Carvalho Ó, Leal A. Unlocking the Secrets of knee Joint unloading: a Systematic Review and Biomechanical Study of the Invasive and Non-Invasive Methods and their Influence on knee Joint Loading. *Rheumato Jun.* 2025;5(3):8. <https://doi.org/10.3390/rheumato5030008>.
- [151] Fernandes NATC, et al. Experimental Validation of Time-Explicit Ultrasound Propagation Models with Sound Diffusivity or Viscous Attenuation in Biological Tissues using COMSOL Multiphysics. *Bioengineering* 2025.